



Molecular dynamics simulations of primary damage formation and tensile properties of irradiated Cu–Ni binary alloys

Xiao-Ping Wei^{a,c,*}, Hairui Li^a, Yaling Zhang^{b,**}, Xue Li^{a,b}, Chen Li^d

^a The School of Mathematics and Physics, Lanzhou Jiaotong University, Lanzhou, 730070, China

^b Institute of Modern Physics, Chinese Academy of Sciences, Lanzhou, 730000, China

^c Gansu Center for Fundamental Research in Complex Systems Analysis and Control, Lanzhou Jiaotong University, Lanzhou, Gansu, 730070, China

^d Advanced Nuclear Energy Team, Advanced Energy Science and Technology Guangdong Laboratory, Huizhou, 516007, China

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ABSTRACT

Cu–Ni alloys are widely used in high-performance applications, such as in nuclear reactors, spacecraft, and other extreme environments, due to their excellent mechanical properties. However, under extreme irradiation conditions, these alloys undergo significant degradation in their microstructure and mechanical properties, which affects their long-term performance. This study investigates the effects of irradiation on the microstructure and mechanical properties of Cu–Ni alloys using molecular dynamics (MD) simulations, focusing on the displacement cascade process under varying temperatures, Primary Knock-on Atom (PKA) energies, crystal orientations, and Ni contents. The results show that higher temperatures facilitate the recombination of interstitial atoms and vacancies. Thereby reducing the number of Frenkel defect pairs in the steady state, while increasing PKA energy significantly increases the number of defect pairs due to more intense cascade collisions. Additionally, the study shows that defects exhibit different behaviors in the [100], [110], and [111] orientations, highlighting the anisotropic nature of Cu–Ni alloys. Tensile simulations indicate that although irradiation reduces the yield strength of Cu–Ni alloys, alloys with higher Ni content exhibit a smaller decrease in yield strength, especially after irradiation, suggesting that increasing Ni content plays a significant role in enhancing the alloy's irradiation resistance and maintaining mechanical performance under irradiation conditions. This study explores the effects of temperature, PKA energy, crystal orientation, and Ni content on irradiation-induced damage in Cu–Ni alloys, aiding the development of radiation-resistant materials and providing a theoretical foundation for improving the mechanical stability and performance of materials used in extreme irradiation environments.

1. Introduction

When materials undergo irradiation, defects such as interstitial atoms, vacancies, and dislocations are generated, altering the material's microstructure [1,2]. The accumulation and evolution of these defects over time can ultimately modify the material's macroscopic properties [3,4]. Among these, the most significant impact is on the mechanical properties. Due to the effects of irradiation, mechanical properties typically degrade, leading to a reduction in ductility, toughness, and fatigue strength [5,6]. Research has shown that irradiation-induced defects interact with pre-existing defects within the material, forming defect clusters, layered tetrahedrons, voids, and other structural configurations. Such interactions may eventually contribute to material

failure [7–9]. Therefore, investigating irradiation damage is crucial for understanding both microstructural changes and macroscopic performance in materials, providing essential insights for designing and optimizing material properties.

Cu–Ni alloys are widely used in material science due to their diverse physical and chemical properties, which vary significantly with composition [10]. The addition of different elements can dramatically influence their mechanical, thermal, and optical properties. For instance, Suemoto et al. investigated the composition-dependent ultrafast luminescence in Cu–Ni alloys, combining experimental techniques and ab initio calculations to reveal the critical role of the Cu–Ni ratio in optical responses [11]. Additionally, Paras and Allanore provided experimental evidence of ordering in Cu–Ni alloys through electronic

* Corresponding author. The School of Mathematics and Physics, Lanzhou Jiaotong University, Lanzhou, 730070, China.

** Corresponding author. Institute of Modern Physics, Chinese Academy of Sciences, Lanzhou, 730000, China.

E-mail addresses: weixp2008@lztu.edu.cn (X.-P. Wei), zhangyl@impcas.ac.cn (Y. Zhang).

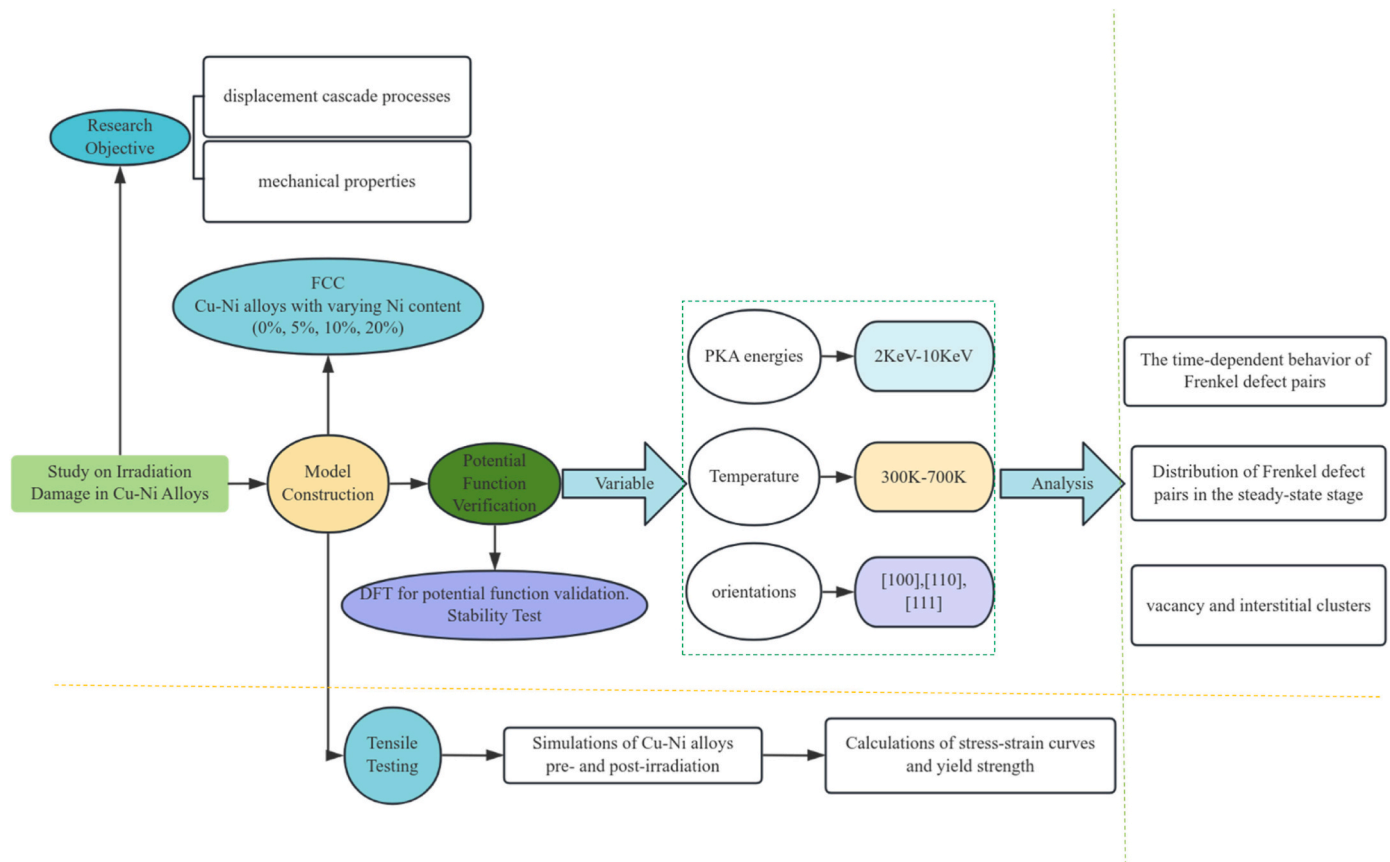


Fig. 1. Detailed flow chart of the simulation framework used in this study.

entropy measurements, further deepening our understanding of their structural properties and behavior at the atomic level [12]. Furthermore, Ak et al. compared experimental results with molecular dynamics (MD) simulations to investigate the mechanical properties of Cu–Ni alloys [13]. Similarly, the incorporation of nanoparticles, as explored by Rachid H B et al., significantly improves crack propagation resistance and other mechanical properties [14], providing valuable insights into how composition affects their tensile strength and other mechanical behaviors. Moreover, Cu and Cu–Ni alloys are esteemed for their superior resistance to corrosion and heat, which further enhances their application potential in various fields [15]. These alloys are commonly utilized in heat-conducting elements in nuclear reactors, radiation shielding materials (including γ /X-rays (10 kV–10 MeV), fast neutrons (0.1–20 MeV), and electrons and β -particles (<2 MeV)), as well as in spacecraft components, such as propulsion systems (combustion chamber bushings and cooling channels of liquid rocket engines), power systems (interconnections and conductive components of satellite solar panels), thermal control systems (heat pipes and radiators), and structural components (spacecraft brackets and fasteners) [16]. The combined understanding of their physical properties, mechanical performance, and diverse applications underscores the importance of Cu–Ni alloys in advancing both material science research and practical engineering applications.

However, under extreme conditions like elevated temperatures and intense irradiation, Cu and Cu–Ni may develop defects that impair their macroscopic properties [17]. Recently, irradiation damage in Cu has been thoroughly investigated using both experimental and theoretical approaches [18,19]. Fujino et al. [20] examined the behavior of single-crystal pure Cu and Cu–Ni alloys under ion beam irradiation using Monte Carlo simulations. Their study demonstrated that the damage depth in Cu samples was significantly greater than in Cu–Ni alloy samples. Wagner et al. [21] investigated how electron radiation affects the

decomposition kinetics and morphological changes in Cu–Ni–Fe alloys. Their experimental observations elucidated the electron beam's impact on internal structural changes and phase separation within the alloy. Poerschke et al. [22] studied the effects of low-temperature (4 K and 200 K) electron irradiation on the decomposition behavior of Cu–Ni alloys during isothermal annealing. Their research demonstrated that electron irradiation induces defects that migrate and cluster during annealing, ultimately causing alloy decomposition. Zinkle et al. [23] investigated the microstructural evolution of Cu–Ni alloys with varying Ni contents under neutron irradiation at 210 °C and 420 °C. It was shown that no void swelling was detected in samples irradiated at 210 °C, instead, a moderate density of dislocations was observed. Plantz et al. [24] examined the influence of ion irradiation-induced dislocation rings on the hardness of Ni–Cu alloys, showing that irradiation leads to a high density of dislocations. Additionally, Poerschke et al. [25] investigated the behavior of interstitial diffusion in Cu–Ni alloys under electron irradiation. They found a strong correlation between the rate of irradiation-induced decomposition and temperature, suggesting an increase in decomposition rate with higher irradiation temperatures. At the lowest irradiation temperature (120K), radiation-induced interstitial migration facilitated interdiffusion in Cu–Ni alloys. Poerschke et al. [26] found that during electron irradiation of Cu–Ni alloys, gap atoms and vacancies diffuse and recombine. Consequently, isothermal irradiation causes alloy decomposition, leading to decreased resistivity. Ryan et al. [27] studied the effects of neutron irradiation on Cu–Ni alloys through the magnetization rate method. Their results showed that neutron irradiation enhances diffusion, increasing magnetization post-irradiation. Scheuer [28] used transmission electron microscopy to examine ion irradiation's effects on the phase stability of Cu–Ni–Fe alloys. They found that heavy ions (Cu, Ne, and N) led to precipitate phase dissolution, while proton irradiation accelerated decomposition and coarsening. Ghauri et al. [29] explored the effects of irradiation

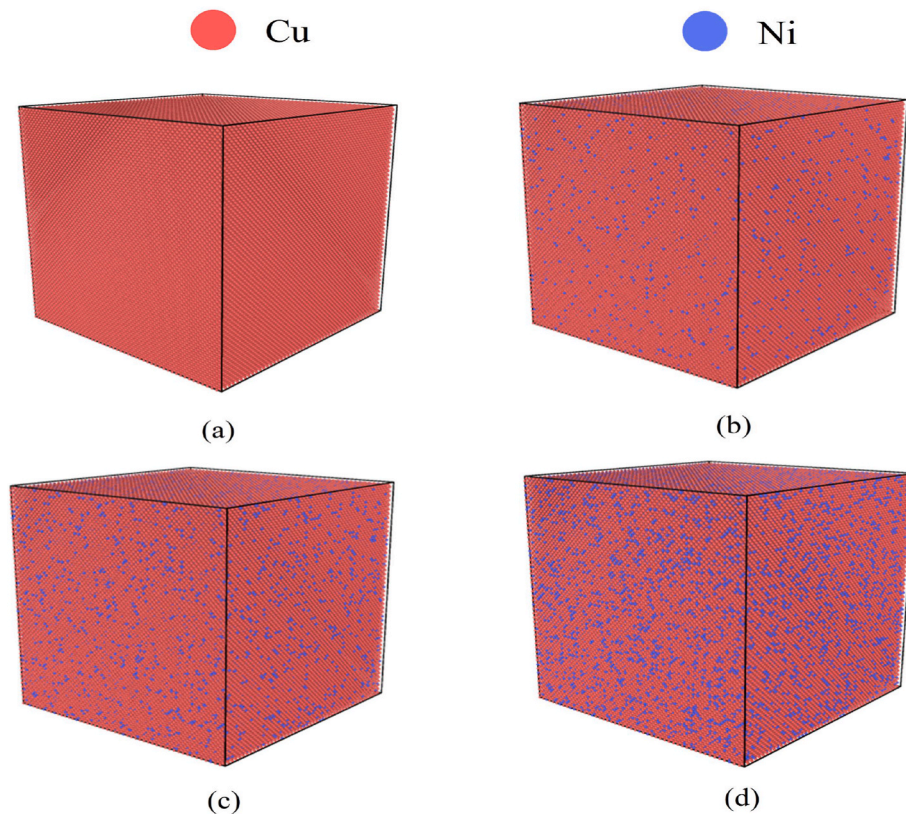


Fig. 2. Structural models of (a) Cu-0at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni.

annealing on the tensile properties of Cu–Ni alloys, showing that irradiation annealing improved ductility while reducing alloy strength. Numerous studies have shown that alloying elements can modulate a metal's hardness, strength, wear resistance, corrosion resistance, and other properties [30–32]. Therefore, studying alloy properties enhances our ability to understand and optimize the performance of metallic materials, aligning them to meet diverse requirements across fields.

In previous studies, there has been a significant gap in understanding the effects of irradiation on the microstructure and mechanical properties of Cu–Ni alloys under various environmental conditions, particularly regarding the mechanical performance of materials post-irradiation. While some studies have explored the impact of irradiation-induced point defects on alloy properties, these studies often focus on limited experiments or theoretical analyses under specific conditions, lacking a comprehensive investigation into the effects of varying temperatures, Primary Knock-on Atom (PKA) energies, and crystal orientations. Therefore, this study innovatively employs molecular dynamics simulations to systematically investigate the effects of irradiation on the displacement cascade process in Cu–Ni alloys under temperatures ranging from 300 K to 700 K, as well as varying PKA energies and crystal orientations. Unlike previous studies, this research not only focuses on irradiation damage at high temperatures (700 K), but also places particular emphasis on irradiation effects across a broader temperature range, from 300 K to 700 K, enabling the findings to be applicable to a wider range of real-world environments, such as marine engineering, nuclear power plants, and spacecraft. By studying this range of temperatures, the study expands the applicability of Cu–Ni alloys, making them suitable for various environmental conditions. Moreover, by considering the coupled effects of temperature, PKA energy, and crystal orientation, the study provides a comprehensive analysis of the irradiation effects on Cu–Ni alloys and explores the irradiation damage mechanisms under different conditions. The study also incorporates tensile simulation testing to analyze the changes in mechanical properties of Cu–Ni alloys before and after irradiation,

revealing the specific effects of irradiation on their strength, ductility, and other mechanical properties. The innovation of this study lies not only in its multidimensional research approach to irradiation damage but also in its systematic evaluation of the mechanical performance, filling critical gaps in the study of irradiation damage under different environmental conditions. The findings offer crucial data for the design of more radiation-resistant materials, further expanding the application potential of Cu–Ni alloys in complex environments, and laying a solid theoretical foundation for the development of radiation-resistant materials in the future.

2. Simulation details

As shown in Fig. 1, this study systematically elucidates the defect evolution behavior of Cu–Ni alloys under irradiation and its impact on macroscopic mechanical properties. The primary objective is to investigate the displacement cascade process during irradiation and the changes in mechanical properties before and after irradiation. To achieve this, supercell models with face-centered cubic structures containing 0, 5, 10, and 20 at.% Ni were constructed. The embedded atom potential was quantitatively validated using density functional theory, and stability tests were performed on the potential function to ensure the reliability of subsequent molecular dynamics simulations. A three-dimensional parameter space was then introduced, covering primary knock-on atom (PKA) energies from 2 to 10 keV, environmental temperatures between 300 and 700 K, and crystal orientations of [100], [110], and [111], in order to encompass the primary variable ranges for irradiation conditions. After completing the displacement cascade simulations, statistical analyses were conducted on the time-dependent Frenkel pair generation-annihilation dynamics, steady-state defect spatial distribution, and the aggregation behavior of vacancy and interstitial atom clusters, providing insights into the microstructural defect evolution mechanism. Simultaneously, quasi-static tensile tests were carried out on the pre- and post-irradiation samples to extract

engineering stress-strain curves and yield strengths, revealing the extent of mechanical property degradation due to irradiation damage. Through the seamless integration and feedback between these stages, this approach achieves a comprehensive fusion of potential function validation, irradiation simulation, and mechanical evaluation, offering a complete computational framework and data support for understanding the structural stability and performance degradation of Cu–Ni alloys under extreme service conditions.

In this paper, in order to validate the effectiveness and accuracy of the potential function FeCuNi-with-ZBL.eam.alloy, the VASP software [33], based on density-functional theory (DFT) is used for the calculation of the first principles. The Kohn-Sham equations are solved using the Generalized Gradient Approximation (GGA) exchange-correlation functional, specifically in the form of Perdew-Burke-Ernzerhof (PBE). The plane-wave truncation energy is set at 450 eV, and the Monkhorst-Pack K-point mesh is $7 \times 7 \times 7$. The molecular dynamics (MD) simulations are conducted utilizing the large-scale atomic-molecular parallel simulation package LAMMPS [34]. Energy minimization is achieved through the conjugate-gradient (CG) algorithm, with a time step of 0.001 ps. The displacement cascade and tensile properties of Cu–Ni alloys at varying temperatures, PKA energies, and orientations have been simulated using the large-scale atomic-molecular parallel simulation package, LAMMPS [34]. The face-centered cubic (fcc) structure of Cu was established, and three distinct Cu–Ni alloys with varying Ni contents were constructed by randomly substituting Ni atoms at the lattice sites of Cu. The displacement cascade process under diverse irradiation conditions was initially simulated for both Cu and Cu–Ni alloys. The interatomic interactions for Cu were modeled using the potential function developed by Mishin et al. [35] and the interatomic interactions for Cu–Ni alloys were described by the potential function proposed by Tramontina et al. [36], which is elucidated through the Ziegler-Biersack-Littmark (ZBL) potential for proximity interactions between atoms. The current interatomic potentials represent a modified version from 2009 by Bonny et al. [37], this potential function is suitable for calculating interatomic interactions in alloy systems. It incorporates short-range ZBL corrections to account for the strong repulsive forces that arise when the interatomic distance is less than 0.5 Å during atomic collisions. By smoothly connecting the ZBL potential with the EAM potential, it effectively describes the repulsive interactions between recoil atoms and target atoms, making it well-suited for simulating the irradiation process of Cu–Ni alloys.

This potential function was also employed in the study presented in Ref. [38], which examined the influence of temperature on the deformation response of fcc Cu–Ni alloys with varying compositions under normal pressure. To mitigate the impact of cell size on the irradiation displacement cascade process and to accommodate irradiation at elevated temperatures and energies, models of adequate size were developed. An initial model of $50a \times 50a \times 50a$, where a is the lattice constant of the alloy ($a = 3.615$ Å), containing a total of 500,000 atoms, as depicted in Fig. 2. The irradiation temperature was regulated using the Nose-Hoover thermal bath method, and the initial model was executed under isothermal and isobaric (NPT) conditions for 20,000 steps with a time step of 0.001 ps, followed by a relaxation period of 20 ps. Subsequently, an atom adjacent to the central atom was designated as the PKA atom, to which a specific velocity was assigned, and cascade collisions were conducted under micro-canonical (NVE) conditions. Each displacement cascade event was simulated for 50 ps and 100 ps, respectively, which is deemed sufficient to dissipate thermal spikes and form stable defects, with the entire cascade adopting periodic boundary conditions.

Irradiation defects were identified through Wigner-Seitz (W-S) analysis [39] using the Open Visualization Tool (OVITO) software [40]. To enhance the accuracy of the results, twenty independent displacement cascade simulations of Cu–Ni alloys were performed at various energies, temperatures, and grain orientations, from which statistical averages were derived.

Tensile properties serve as critical indicators for assessing the impact of radiation damage on materials. Consequently, an analysis of the tensile properties of Cu–Ni alloys, both pre- and post-irradiation, was conducted through the computation of stress-strain curves. The dimensions of the simulated box we prepared are $50a \times 50a \times 50a$. The crystal orientations of the boxes were aligned along the [100], [110], and [111] directions, encompassing a total of 500,000 atoms of the Cu–Ni alloy. Utilizing the embedded atom method (EAM) as proposed by B. Onat et al. [41], the potential function of the semi-empirical many-body model for the Cu–Ni alloy was employed to characterize atomic interactions. The conjugate-gradient (CG) algorithm was implemented for energy minimization, while the Nose-Hoover hot bath method was utilized to maintain the model at a temperature of 300 K. The model underwent relaxation under the NPT ensemble with a time step of 0.001 ps over a total duration of 20 ps, all while maintaining zero pressure in all directions to achieve equilibrium. The stretching process was conducted along the xyz triaxial direction under the NVE ensemble, with a strain rate of 0.005 ps^{-1} . Subsequently, tensile simulations of specimens subjected to irradiation damage were performed to evaluate the influence of such damage on the tensile properties of Cu–Ni alloys. Furthermore, dislocation analysis (DXA) and common neighbor analysis (CNA) [42] were employed in OVITO [40] during the tensile process to investigate alterations in the material's microstructure resulting from stretching.

In this study, a series of key parameters were selected to accurately simulate the behavior of Cu–Ni alloys under irradiation conditions. The temperature range was set from 300 K to 700 K, encompassing typical application temperatures for Cu–Ni alloys in nuclear reactors, aerospace, and high-temperature industrial environments. 300 K represents ambient conditions, while 700 K approaches the maximum temperature these materials may experience in high-radiation environments, particularly in nuclear reactors where materials are exposed to elevated temperatures under radiation. This temperature range allows for the investigation of defect behavior induced by irradiation at different temperatures, particularly the evolution of defects under high temperatures and their long-term impact on material properties. The Primary Knock-on Atom (PKA) energy range was chosen to be 2 keV–10 keV, which covers common irradiation particle energies. The 2 keV–10 keV PKA energy range effectively simulates the initial stage of radiation damage in real-world irradiation environments, producing moderate defect sizes that are ideal for observing how defects evolve over time. The upper limit of 10 keV ensures the comprehensive study of irradiation effects, particularly the defect generation, migration, and aggregation processes under high-energy collisions, while avoiding excessive damage or defect generation caused by overly high energies. This provides a better understanding of Cu–Ni alloy behavior under various irradiation conditions from a practical engineering perspective. Furthermore, Cu–Ni alloys, as face-centered cubic (FCC) crystal structures, exhibit significant anisotropy, and the crystal orientation plays a crucial role in determining the material's microstructure, mechanical properties, and response to irradiation damage. To comprehensively analyze the behavior of Cu–Ni alloys under irradiation, the study selected three crystal orientations: [100], [110], and [111], aiming to investigate the differential effects of irradiation on dislocation movement, defect migration, and aggregation across these orientations. For example, the [100] direction typically exhibits lower dislocation slip resistance, while the [111] direction is more prone to dislocation accumulation. By studying the irradiation effects along these orientations, the research provides insights into the alloy's performance in real irradiation environments, particularly in terms of stress concentration, defect formation, and migration. Lastly, the study selected four Cu–Ni alloy compositions with varying nickel content (Cu-0at%Ni, Cu-5at%Ni, Cu-10at%Ni, and Cu-20at%Ni), with Cu-0at%Ni serving as the baseline. Higher nickel content alloys were included to assess how the addition of nickel affects the alloy's response to irradiation. By examining alloys from pure copper (Cu-0at%Ni) to higher nickel concentrations (Cu-20at

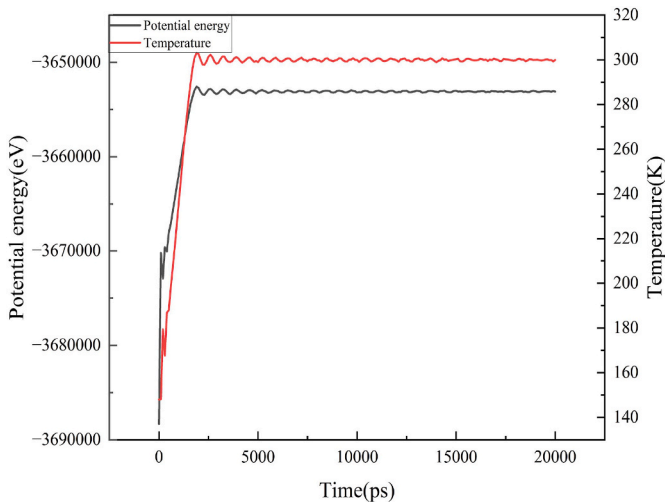


Fig. 3. Variations in the temperature and potential energy of the simulation system during the relaxation period.

%Ni), the study captures the trend of performance enhancement with increasing nickel content, especially in terms of improved resistance to irradiation-induced defects. In summary, the selection of these parameters is justified as they comprehensively reflect the microstructural changes, radiation damage evolution, and mechanical property degradation of Cu–Ni alloys under irradiation, providing a solid theoretical basis for the future application of Cu–Ni alloys in high-temperature irradiation environments.

3. Results

3.1. Potential function validation

To evaluate the applicability and accuracy of the selected interatomic potential (FeCuNi-with-ZBL.eam.alloy) for simulating irradiation damage in Cu and Cu–Ni alloys, a thermodynamic stability test was first performed through a relaxation process. As shown in Fig. 3, both the temperature and potential energy of the system rapidly converged and stabilized within a short period, reaching thermal and energetic equilibrium within approximately 5 ps and remaining stable thereafter. This behavior demonstrates the strong dynamic stability and numerical robustness of the potential under the current model configuration and boundary conditions, thereby providing a solid physical foundation for subsequent irradiation and mechanical behavior simulations.

Building upon this, a series of key physical properties—including the lattice constant, elastic constants, bulk modulus, vacancy formation energy, cohesive energy, and displacement threshold energy—were calculated for both Cu and Cu–Ni alloys using molecular dynamics (MD)

and density functional theory (DFT) methods to comprehensively assess the accuracy of the selected potential function. As shown in Table 1, for pure Cu, the results obtained from MD and DFT showed excellent agreement with available experimental data [43–46] and previously reported literature values [47–49], indicating that the potential function can reliably capture the fundamental physical properties of the material. In addition, comparative calculations were performed for Cu–Ni alloys with varying Ni concentrations, and the MD and DFT results remained consistently aligned. Specifically, for the Cu–20 %Ni alloy, the computed values were further compared with existing reference data, demonstrating a similarly high level of consistency, which reinforces the validity and transferability of the selected potential function across various alloy configurations.

3.2. Displacement cascade process

3.2.1. Effect of temperature on cascade damage

Considering the practical applications of Cu–Ni alloys and to mitigate the influence of sample size on irradiation effects [51], the simulation temperature was established within the range of 300 K–700 K.

To evaluate the influence of temperature on the displacement cascade in Cu–Ni alloys, a PKA energy of 4 keV was applied in the [100] direction (with a deviation angle of 10°). As shown in Fig. 4, the number of Frenkel defect pairs exhibited a consistent trend across different temperatures: the number initially increases sharply, reaches a peak, then gradually decreases, and ultimately stabilizes. For the Cu–Ni alloy system, with increasing temperature, the number of Frenkel defect pairs at the peak increases. This is attributed to the enhanced atomic kinetic energy at higher temperatures, which promotes the generation, accumulation, and retention of more defects, resulting in an increase in defect pairs at the peak stage.

As shown in Fig. 5, during the stabilization phase, the number of Frenkel defect pairs decreases with increasing temperature, but the defects become more localized. This suggests that higher temperatures facilitate atomic motion, allowing defects to move more freely and recombine. The increase in temperature accelerates the recombination of vacancies and interstitial atoms, thereby reducing the number of defects in the stabilization phase. The higher thermal energy enhances atomic mobility, leading to a more compact distribution of defects and a reduction in the number of large defect clusters.

To elucidate the effect of varying Ni contents on Frenkel defect pairs at the same temperature, we examined the influence of different Ni concentrations on the defect pair count at 300 K (see Fig. 6). Combined with the analysis of Fig. 4, it is evident that higher Ni content results in fewer defects at the peak stage compared to pure Cu. This indicates that the incorporation of Ni alters the microstructure of the alloy, potentially by promoting the formation of other phases (such as BCC and HCP), which reduces defect formation and enhances defect recombination efficiency. These results highlight the significant impact of both temperature and Ni content on the defect evolution of Cu–Ni alloys under

Table 1

Based on density functional theory (DFT) and molecular dynamics simulation (MD), the lattice constant a (Å), elastic constant C_{ij} (GPa), volume modulus B_m (GPa), vacancy formation energy E_v (eV), cohesion energy E_c (eV), and displacement threshold E_d (eV) of Cu and Cu–Ni alloys were calculated respectively. The calculation methods of Cu were compared using the existing experimental data and DFT data.

Alloys	Method	a	C_{11}	C_{22}	C_{44}	B_m	E_v	E_c	E_d
Cu	exp.	3.615 [43]	176.20 [43]	124.94 [43]	81.77 [43]	142.03 [43]	1.28 [44]	–3.52 [45]	30 [46]
	DFT	3.640 [47]	171.10 [47]	122.20 [47]	75.30 [47]	138.50 [47]	1.33 [48]	–3.32 [49]	
	DFT	3.629	181.57	121.36	74.28	141.43	1.31	–3.41	
	MD	3.631	171.25	123.93	76.45	139.70	1.29	–3.50	26
Cu–5 %Ni	DFT	3.629	179.86	125.48	74.28	143.16	1.28	–3.45	29
	MD	3.633	170.09	121.77	76.40	137.33	1.27	–3.54	
Cu–10 %Ni	DFT	3.623	183.11	123.49	84.63	143.36	1.25	–3.52	31
	MD	3.617	170.74	120.63	76.44	137.33	1.24	–3.58	
Cu–20 %Ni	DFT	3.610	187.84	129.51	87.04	148.95	1.19	–3.57	32
	DFT	3.587 [41]	186.9 [50]	131.1 [50]	87.8 [50]	146.7 [50]	1.31 [41]	–3.69 [41]	
	MD	3.584	170.09	117.77	75.91	135.57	1.21	–3.66	

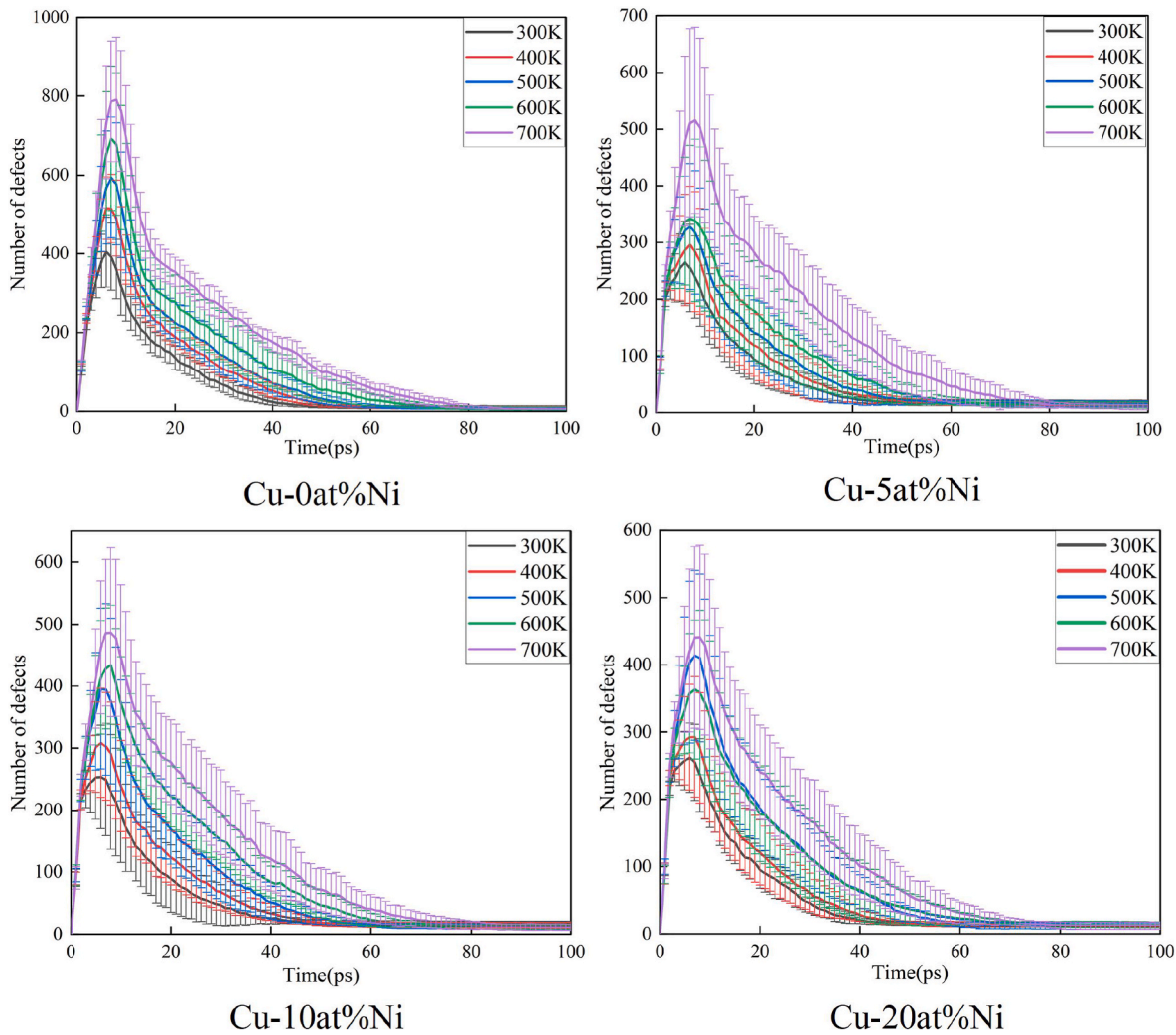


Fig. 4. The change of the Frenkel defect logarithm of (a) Cu- 0 at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni in the direction [100] with time at different temperatures and at PKA energy of 4 keV.

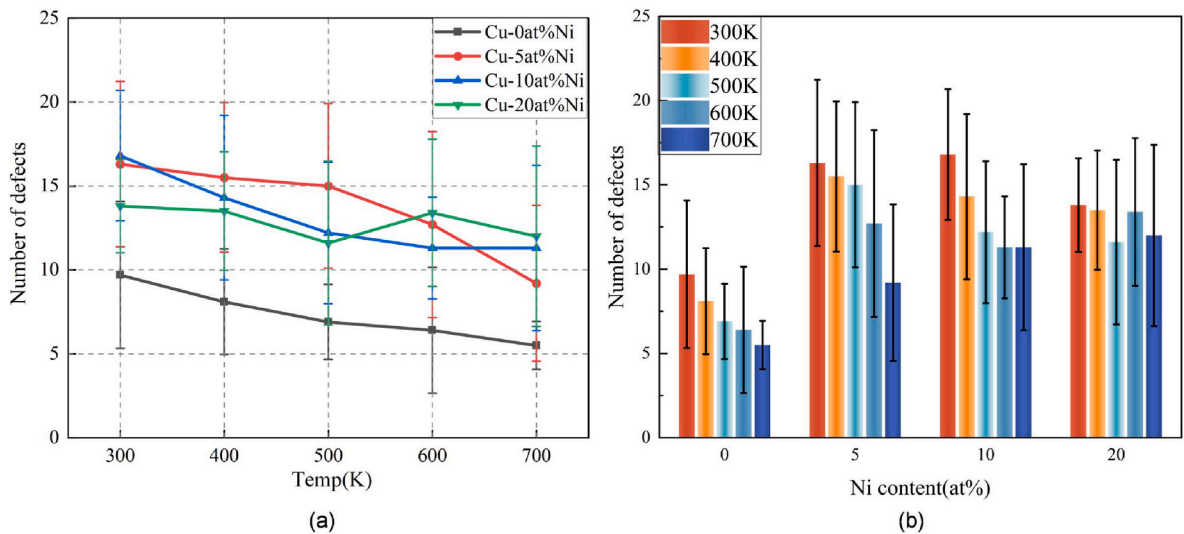


Fig. 5. Number of Frenkel defect pairs in Cu–Ni alloys during the stable stage at different temperatures. (a) Defect count as a function of temperature for Cu alloys with varying Ni contents. (b) Defect count at different temperatures for Cu–Ni alloys with varying Ni contents.

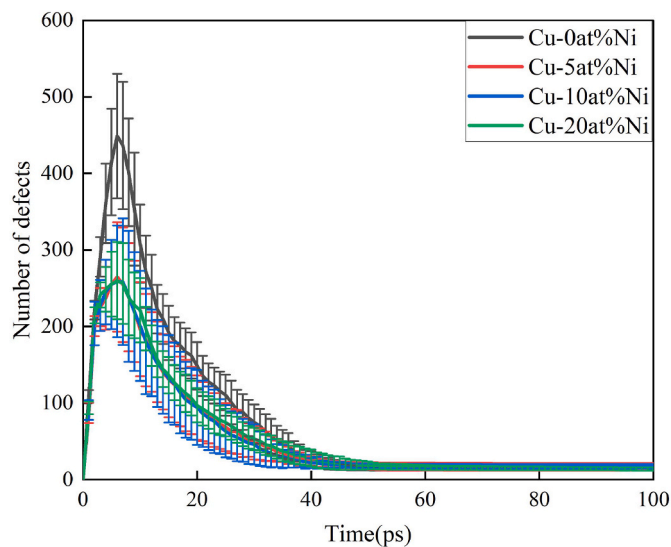


Fig. 6. The number of Frenkel defect pairs in Cu and Cu–Ni alloys changes over time with PKA energy of 4 keV along the [100] direction at 300 K.

irradiation, suggesting that the generation and recombination processes of defects play a decisive role in the material's radiation damage behavior in high-temperature environments.

Fig. 7 illustrates the final distribution of Frenkel defect pairs in the Cu-10at%Ni alloy at different temperatures. As the temperature increases, interstitial atoms are primarily located around vacancies, with a small number of interstitial atoms and vacancies clustering to form interstitial and vacancy clusters. This further confirms that temperature has a significant impact on both the quantity and distribution of defect pairs. The increase in temperature intensifies atomic thermal motion, facilitating the approach and recombination of interstitial atoms and vacancies, leading to a reduction in the number of defect pairs and a

more localized defect distribution. This phenomenon indicates that, at higher temperatures, defects are more likely to merge and recombine, forming more compact defect clusters and reducing the number of independent defect pairs in the system. The localized defect distribution suggests that high temperatures not only affect the number of defect pairs but also alter their spatial distribution, further influencing the alloy's microstructural evolution and macroscopic mechanical properties.

3.2.2. Effect of PKA energy on cascade damage

To study the influence of PKA energy on the displacement cascade in Cu–Ni alloys, we conducted experiments at a temperature of 300 K. The PKA energy was changed in the [100] direction (deviation angle 10°), from 2 keV to 10 keV, with an increment of 2 keV. The temporal evolution of Frenkel defect pairs in Cu–Ni alloys at different PKA energies is illustrated in Fig. 8. The results show that the number of Frenkel defect pairs produced by atomic cascade collisions in Cu–Ni alloys increases significantly with the increase in PKA energy. These observations align with previous research conducted on W [52] and Fe, Cu, and Zr.

Fig. 9 presents the correlation between the number of Frenkel defect pairs and the Ni content ratio in the stable structural configuration at 300 K for varying PKA energies. At PKA energies of 2 keV and 4 keV, the number of Frenkel defect pairs in the system exhibits a gradual increase with rising Ni content when the Ni content ratio is 5 at% and 10 at%, but decreases when the Ni content ratio reaches 20 at%. At PKA energies of 6 keV and 10 keV, the number of defect pairs diminishes as the Ni content increases. At 8 keV, the number of defect pairs decreases with Ni content at 5 at% and 10 at%, but increases at 20 at%. Fig. 10 illustrates the displacement cascade process in Cu–Ni alloys at 300 K under a PKA energy of 10 keV in the [100] direction.

Fig. 10(a) depicts the initial defect generation stage, Fig. 10(b) represents the thermal peak stage, and Fig. 10(c) shows the stabilization phase. From Fig. 10(a) and (b), it is apparent that defects are generated rapidly through cascade collisions, reaching a peak in the number of vacancies and interstitial atoms. Subsequently, these defects recombine, with the majority disappearing and stabilizing around 50 ps, as shown in

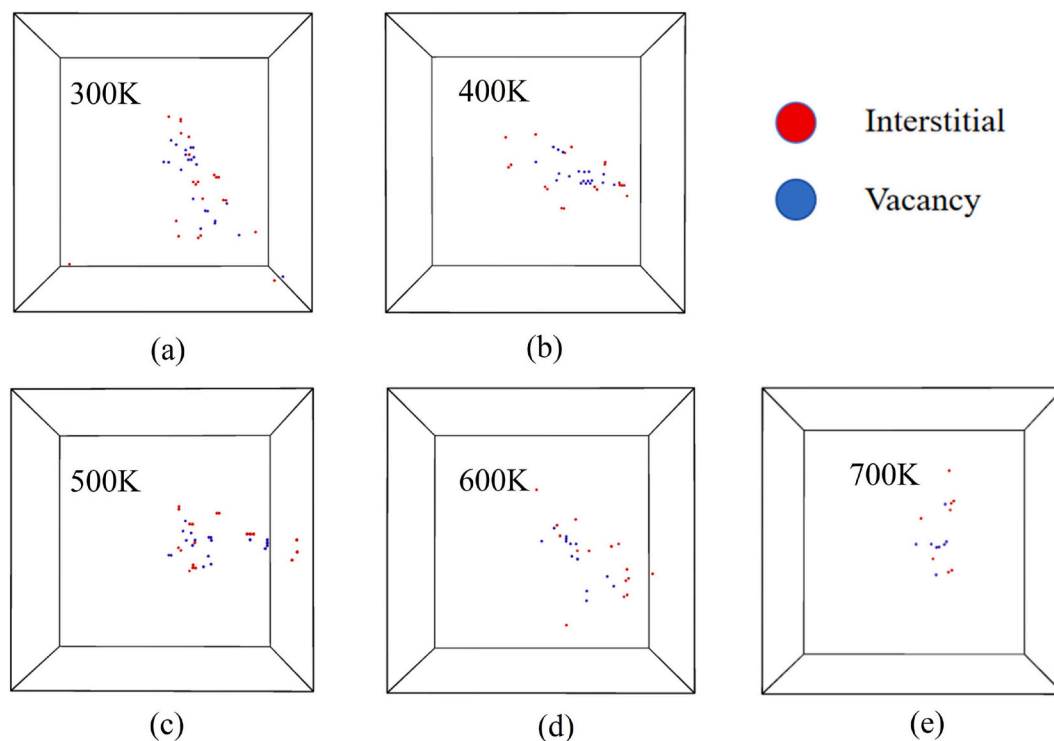


Fig. 7. The final distribution of Frenkel defect pairs ($t = 100$ ps) in Cu-10at%Ni alloy at different temperatures with PKA energy of 4 keV and along the [100] direction, where red and blue circles represent interstitial atoms and vacancies, respectively.

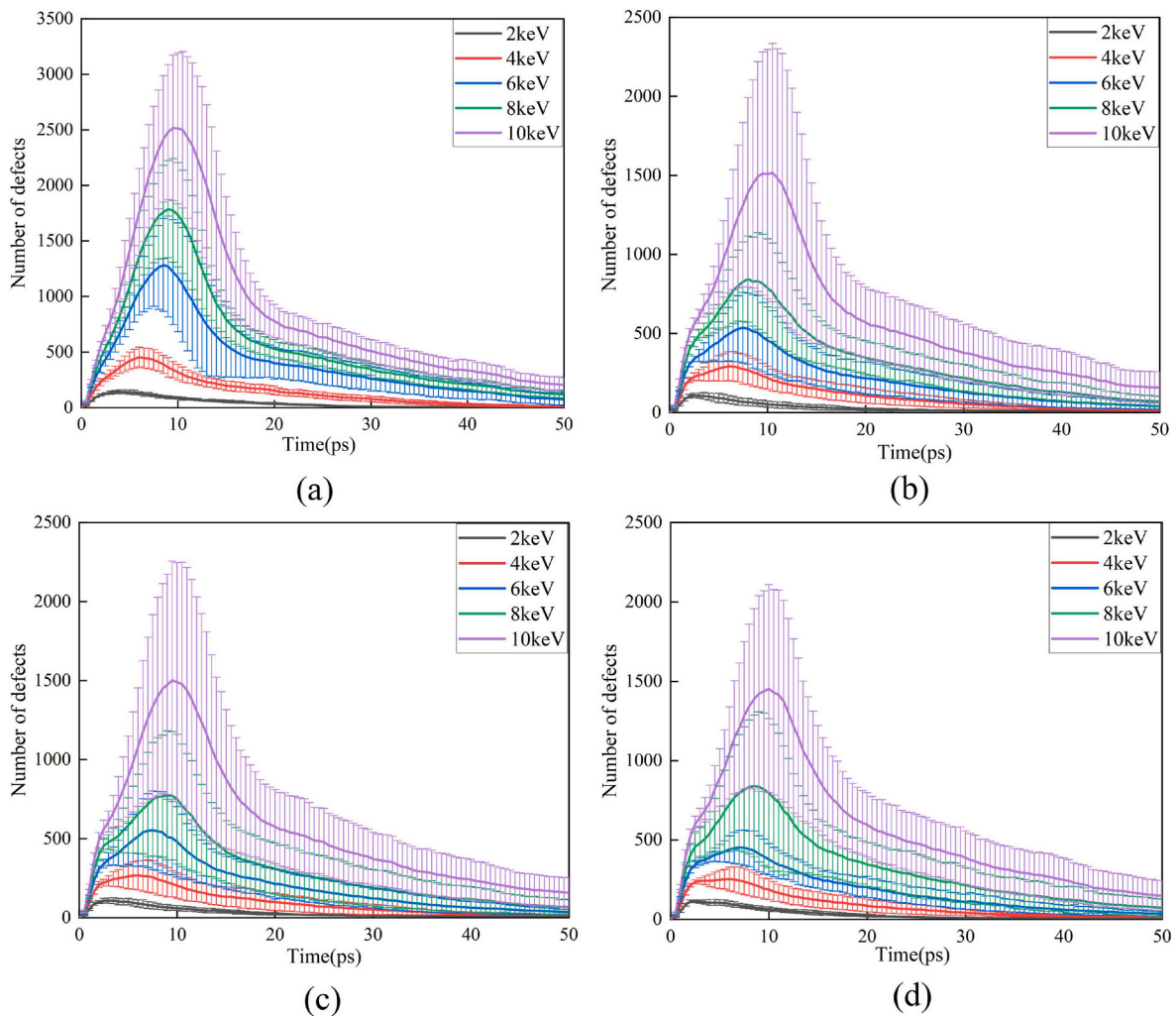


Fig. 8. The number of Frenkel defect pairs of (a) Cu-0at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni under different PKA energy irradiation changes with time at 300 K.

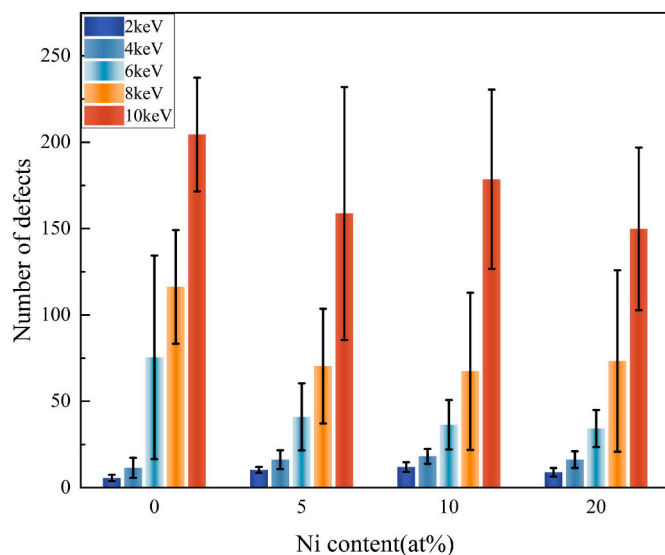


Fig. 9. The number of Frenkel defect pair($t = 50$ ps) as a function of Ni content with different PKA energies at 300 K.

[Fig. 10\(c\)](#). For the same system, the number of Frenkel defect pairs in the final stabilized structure increases with higher PKA energy, resulting in a more localized distribution of defects.

The size of defect clusters resulting from energetic particle collision cascades is a critical parameter in characterizing the microstructure of displacement cascades in irradiated materials [53]. Defect clusters are quantified by the number of point defects they encompass, with their size being directly proportional to this count. The identification of these clusters is facilitated by a truncation radius, which is determined by selecting the third nearest neighbor for interstitial atoms and the second nearest neighbor for vacancies. Consequently, an interstitial atom clusters with atoms located within its third nearest neighbor position, while a vacancy clusters with atoms within its second nearest neighbor [54]. [Fig. 11](#) illustrates the defect clusters in Cu–Ni alloys at various PKA energies at 300 K, with [Fig. 11\(a\)](#) displaying vacancy clusters and [Fig. 11\(b\)](#) displaying interstitial clusters. As depicted in [Fig. 11](#), for both the Cu and Cu–Ni systems, the number of vacant and interstitial clusters formed increases with the elevation of PKA energy, while simultaneously decreasing with an increase in cluster size. Notably, the predominant cluster size observed is 1 Å clusters. Whether it is the vacancy clusters or interstitial clusters, all types exhibit a decrease in number with increasing Ni content in the system. Furthermore, interstitial clusters demonstrate a greater propensity to aggregate and form larger clusters compared to vacancy clusters, a finding that is consistent with previous reports [55–57].

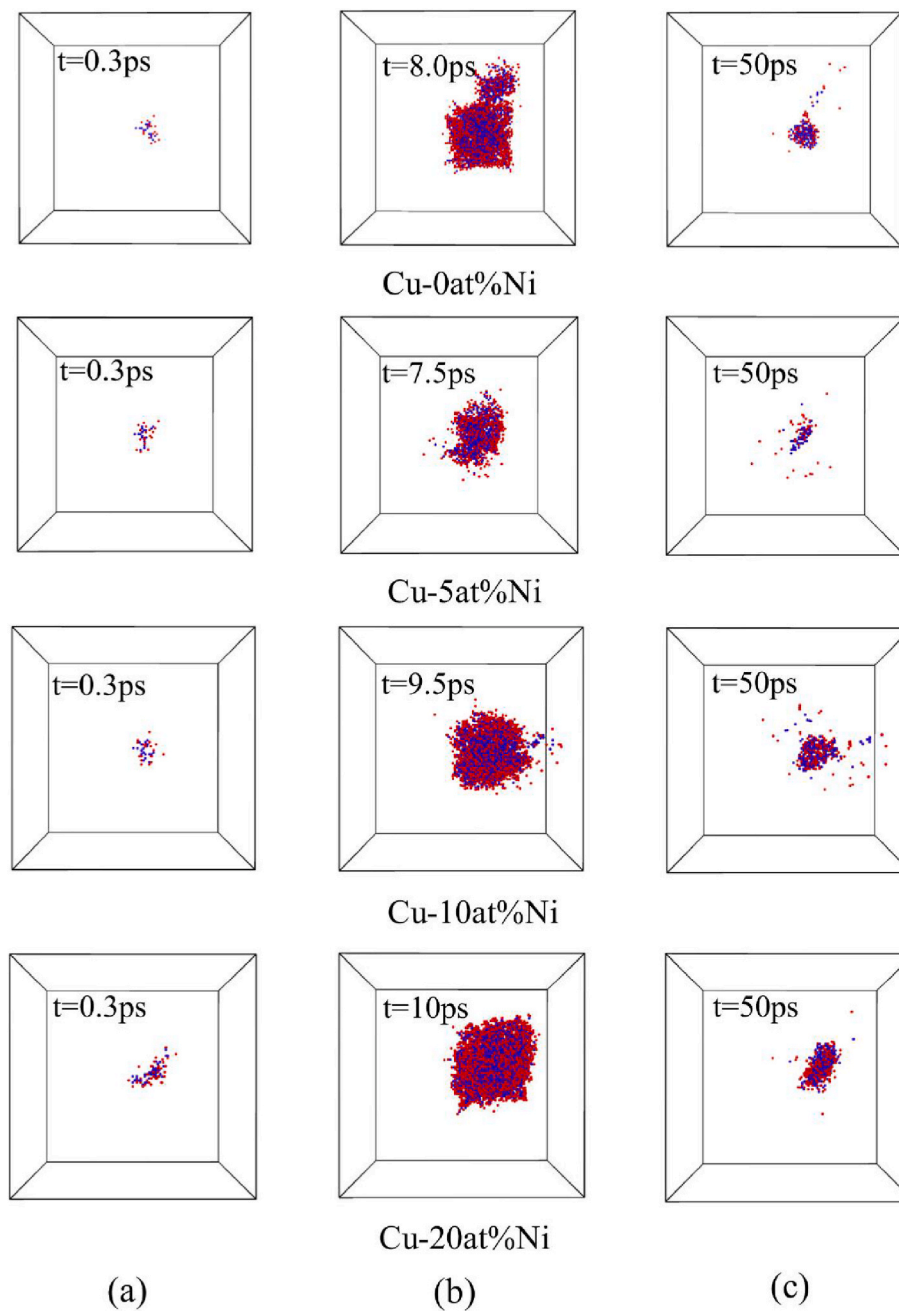


Fig. 10. The number of Frenkel defect pairs in Cu and Cu–Ni alloys at 300 K with a PKA energy of 10 keV along the [100] direction, (a) initial stage, (b) peak stage and (c) stable stage, where red and blue circles represent interstitial atoms and vacancies, respectively.

3.2.3. Effect of PKA direction on cascade damage

In a subsequent analysis, we investigated the effect of PKA atoms with different crystal orientations on the displacement cascade. Specifically, we focus on the effect of orientation on displacement cascades along [100], [110] and [111] at a temperature of 300 K and a PKA energy of 4 keV. The temporal variation in the number of Frenkel defect pairs in Cu–Ni alloys for different PKA orientations is depicted in Fig. 12.

Notably, in Fig. 12(a) Cu system, the highest number of Frenkel defect pairs is observed in the [111] direction, while the lowest count is recorded in the [100] direction during the peak phase. For systems Fig. 12(b) 5 at% and Fig. 12(c) 10 at% Ni, the maximum number of Frenkel defect pairs is similarly generated in the [111] direction during the peak phase, with the counts in the [110] and [100] directions being nearly equivalent. In Fig. 12(d) 20 at% Ni system, the highest number of Frenkel defect pairs is produced in the [110] direction during the peak

phase, with nearly equal quantities observed in the [111] and [100] directions. Fig. 13 illustrates the final distribution of the number of Frenkel defect pairs for a Cu-10at%Ni alloy at 300 K and 4 keV across different PKA orientations. From Fig. 13, it is evident that interstitial atoms predominantly cluster around vacancies, while a limited number of interstitial atoms and vacancies aggregate to form interstitial and vacancy clusters.

3.3. Tensile properties of Cu–Ni alloys

Temperature, pressure, irradiation, and other factors can cause changes in the microstructure of materials, which in turn affect their macroscopic properties. As Dak G et al. studied the tensile properties of metals at high temperatures by varying pressure [58], their work provides valuable insight for studying the tensile properties of materials by

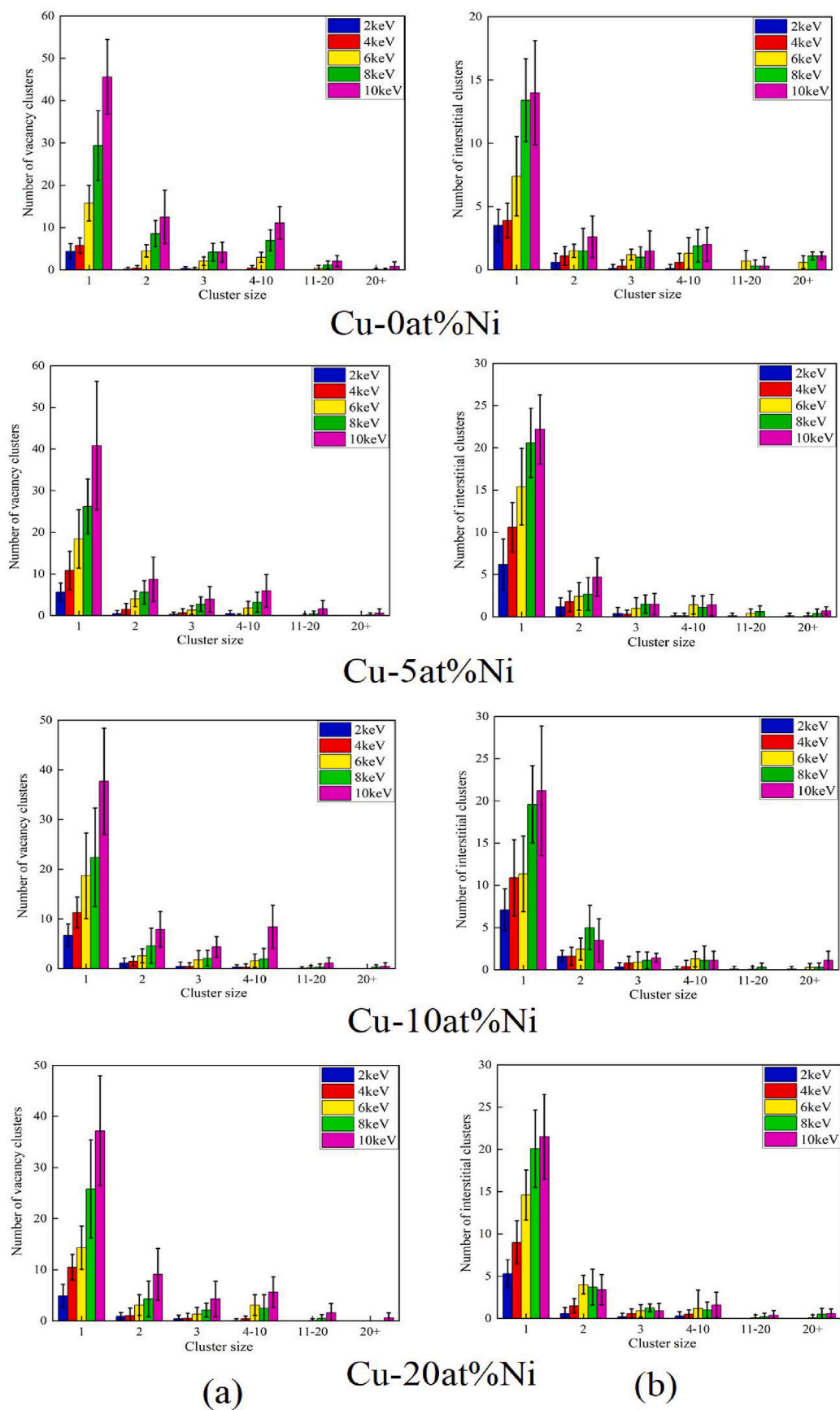


Fig. 11. (a) Vacant clusters and (b) interstitial clusters in Cu and Cu-Ni alloys during displacement damage process with different PKA energies along the [100] direction at 300 K.

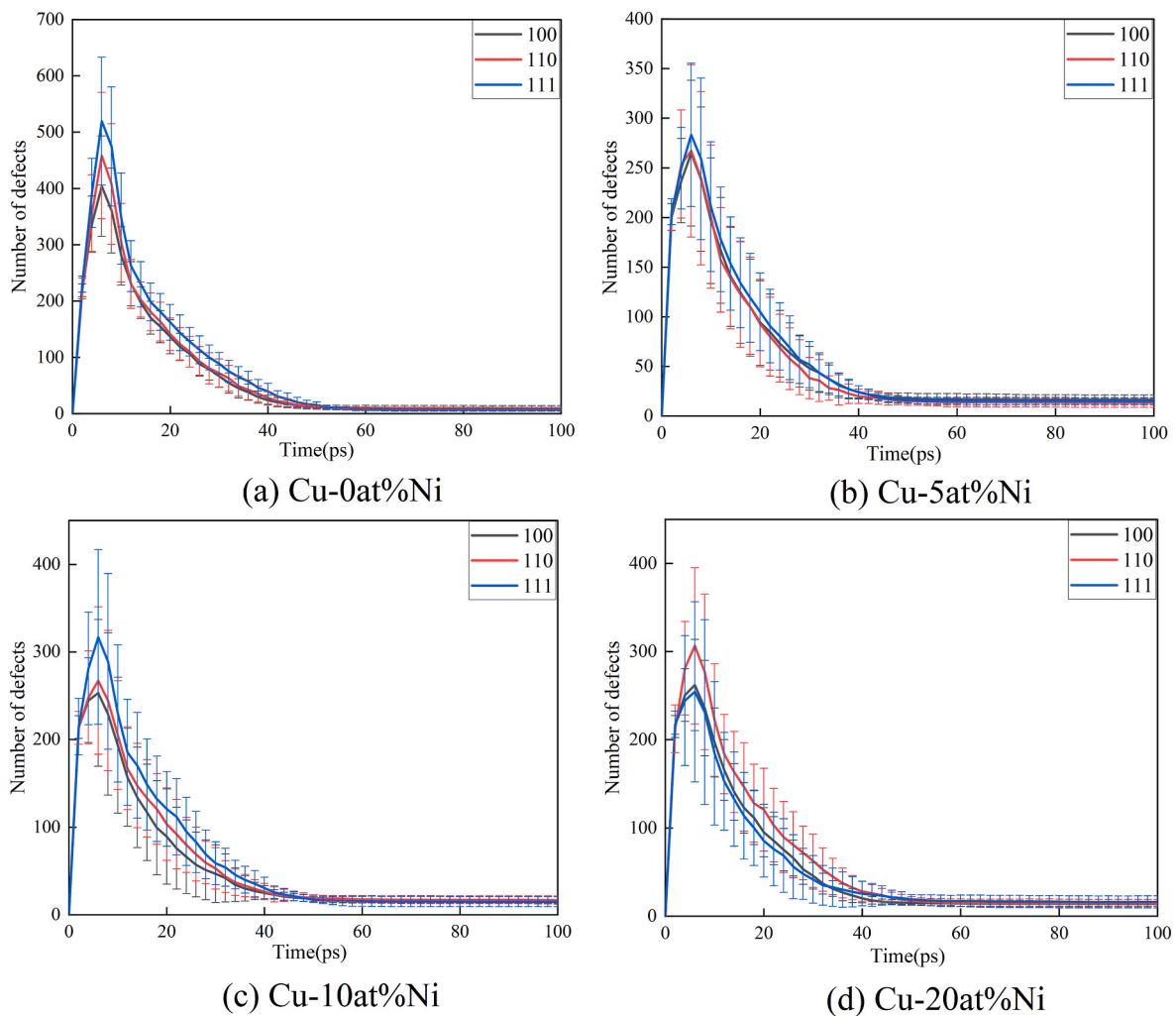


Fig. 12. The number of Frenkel defects in (a) Cu-0at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni varies with time when the PKA energy in different directions is 4 keV at 300 K.

changing irradiation energy. In this section, we primarily focus on the tensile properties of irradiated materials to gain a deeper understanding of the effect of microstructural changes on the mechanical properties of materials [59]. Therefore, we analyzed the mechanical properties of Cu–Ni alloys before and after irradiation by calculating the stress-strain curves.

3.3.1. Tensile properties of Cu–Ni alloys

The incorporation of a second phase element into a metallic matrix is known to modify the mechanical properties of the material [60,61]. The proportion of the second phase element added also affects the alloy's properties, as demonstrated in the study by Maurya A K et al. They investigated the effect of different proportions of the second phase element on the tribological performance of sDSS 2507/IN-625 dissimilar welded joints for marine applications [62]. The study shows that the proportion of the second phase element plays a significant role in altering the alloy's properties, particularly in enhancing wear resistance and corrosion resistance. In this context, the tensile properties of Cu–Ni alloys with nickel contents of 5 at%, 10 at% and 20 at% were investigated.

The results indicated a significant increase in the tensile strength of Cu–Ni alloys with higher Ni content, while the elastic modulus remained relatively constant. Fig. 14 illustrates the stress-strain curves for both Cu and Cu–Ni alloys at a temperature of 300 K. As can be seen in Fig. 14(a), in the strain range of less than 0.025, the stress and strain have a linear

relationship, which is manifested as the elastic deformation stage. As strain continues to increase until the stress reaches its maximum (the yield strength), this phase is referred to as the yielding stage. Subsequent increments in strain result in a marked decrease in stress, indicating the onset of local deformation. Ultimately, as strain continues to increase, the stress tends to stabilize, signaling the gradual failure of the material. Fig. 14(b) provides an enlarged view of the Cu–Ni alloys in proximity to the yield strength, demonstrating that the yield strength of Cu–Ni alloys is significantly greater than that of pure Cu system, with a progressive increase observed as the Ni content in the Cu–Ni alloys. The main reason is that the addition of Ni can produce a solid solution strengthening effect. A solid solution is formed between Ni and Cu atoms, which increases the strength and stability of the lattice and can enhance the strength and hardness of the alloy. Moreover, there are significant differences in the atomic radii of Cu and Ni atoms, which leads to the distortion and deformation of the solid solution lattice. More grain boundaries, dislocations and twinning interfaces are formed inside the alloy crystal, thereby increasing the obstacles that hinder the sliding of dislocations and the movement of grain boundaries. This makes the alloy have higher toughness and durability. Fig. 15 specifically depicts the variation in yield strength of Cu–Ni alloys as a function of Ni content. For example, the yield strength of the Cu alloy is measured at 16.88 GPa, in contrast to 18.99 GPa, 19.27 GPa, and 19.77 GPa for the Cu-5at%Ni, Cu-10at%Ni and Cu-20at%Ni systems, respectively. This data confirms that the yield strength of Cu–Ni systems increases progressively with

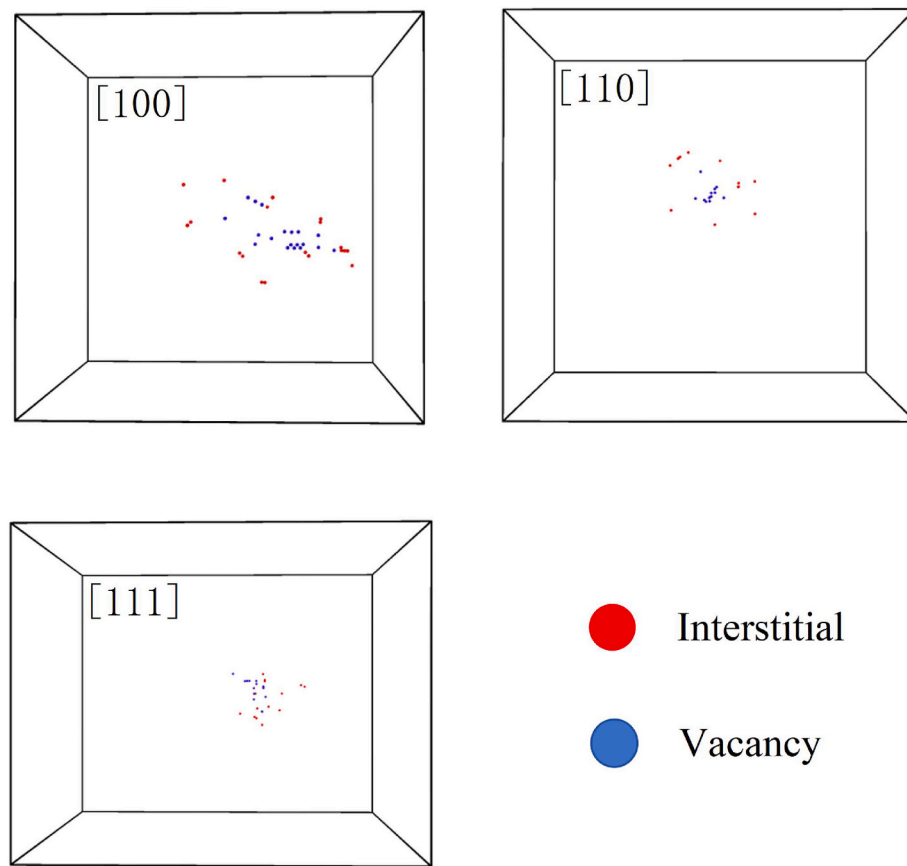


Fig. 13. The final distribution of the number of Frenkel defects ($t = 100\text{ps}$) in Cu-10at%Ni alloy with PKA energy of 4 keV at 300 K and along different directions, where red and blue circles represent interstitial atoms and vacancies, respectively.

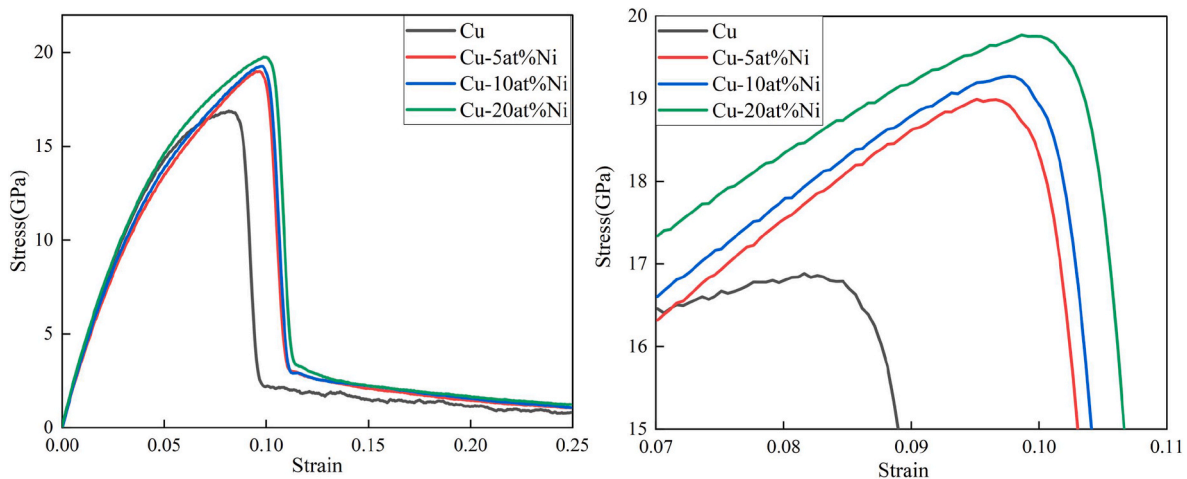


Fig. 14. (a) Stress-strain curves and (b) enlarged stress-strain curves around yield strength for Cu and Cu–Ni alloys at 300 K.

higher nickel content.

The stretching process induces modifications in the microstructure of Cu–Ni alloys, which may lead to structural phase transitions or the formation of dislocations. To investigate these alterations, we utilized the visualization software OVITO. The phase transitions observed during the stretching process are illustrated in Figs. 14 and 15, while the dislocations are depicted in Fig. 18. As shown in Fig. 16, the stretching process led to the conversion of the FCC structures of Cu and Cu–Ni alloys into BCC, HCP, and other novel phase structures.

The incorporation of Ni into the Cu alloy increased the proportion of

the BCC phase within the system. Fig. 17 delineates the trend of phase constituents in Cu–Ni alloys during strain stretching, indicating a decrease in Fig. 17(a) FCC phase and an increase in Fig. 17(b) BCC, Fig. 17(c) HCP, and Fig. 17(d) other phases with the increase of Ni content. From Fig. 18, we identified five distinct types of dislocations: $1/2\langle 110 \rangle$ (Perfect), $1/3\langle 100 \rangle$ (Hirth), $1/3\langle 111 \rangle$ (Frank), $1/6\langle 110 \rangle$ (Strain-rod), and $1/6\langle 112 \rangle$ (Shockley). Notably, the Frank and Shockley dislocations are classified as line dislocations, with the $1/6\langle 112 \rangle$ (Shockley) dislocation line playing a significant role, as its length increases substantially over time, while the lengths of the other

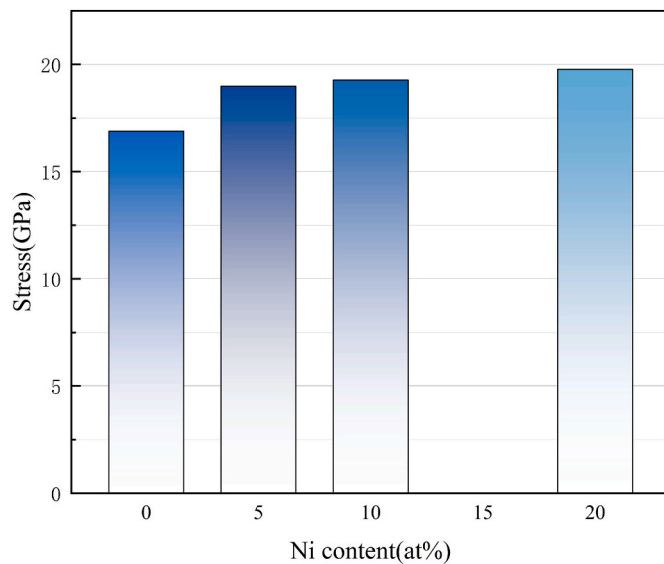


Fig. 15. The yield strength of Cu–Ni alloys changes with Ni content at 300 K.

dislocation lines remain relatively constant.

Fig. 19 presents a schematic of the phase structure and dislocations in Cu–Ni alloys under varying strains. It can be observed that during the

tensile process of Cu–Ni alloys, distinct phase transformations and dislocation evolution characteristics emerge with the accumulation of strain: in the elastic stage (strain = 0.02), the face-centered cubic (FCC) matrix remains stable, with almost no new phases or additional dislocations appearing. As the strain increases to the yield stage (strain = 0.09), local regions begin to generate new phases such as body-centered cubic (BCC) and hexagonal close-packed (HCP), as well as a dislocation network dominated by $1/6\langle 112 \rangle$ (Shockley) dislocations. The "other phases" in question are typically phases that occur in very small fractions or under specific conditions during the deformation process, such as specific intermediate or metastable phases. At a strain of approximately 0.25, FCC still dominates, but a significant accumulation of Shockley dislocations induces prominent stacking faults, which significantly affect plasticity. The origin of phase transformation lies in the instability of the FCC lattice under high strain: atomic rearrangements generate BCC/HCP structures that better accommodate strain to release internal stresses, while dislocation accumulation and slip accelerate this process, especially at high strain rates, where FCC tends to undergo local disorder or transform into BCC. Similar trends are commonly observed in molecular dynamics (MD) tensile simulations [63,64]. However, in experiments, Cu–Ni alloys largely maintain a stable FCC structure even under mechanical loading, due to the following reasons: (1) In MD simulations, the strain rate is kept constant for controlled analysis of phase evolution and dislocation behavior, while in real-world experiments, stress and strain rates vary due to testing conditions, such as loading speed and material preparation, and temperature conditions are often less

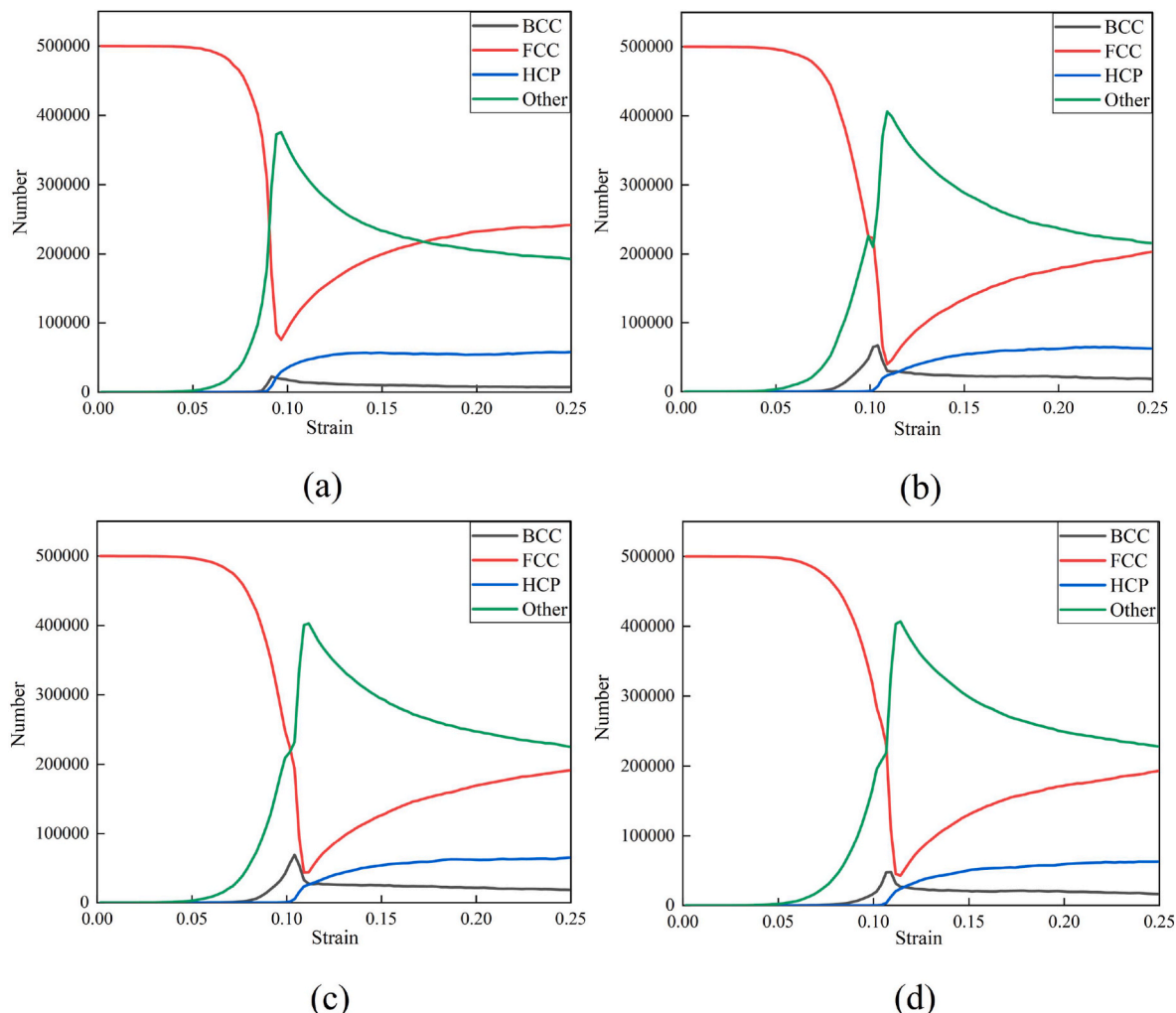


Fig. 16. Structural phase transitions observed in (a) Cu–0 at%Ni, (b) Cu–5at%Ni, (c) Cu–10at%Ni and (d) Cu–20at%Ni during tensile deformation.

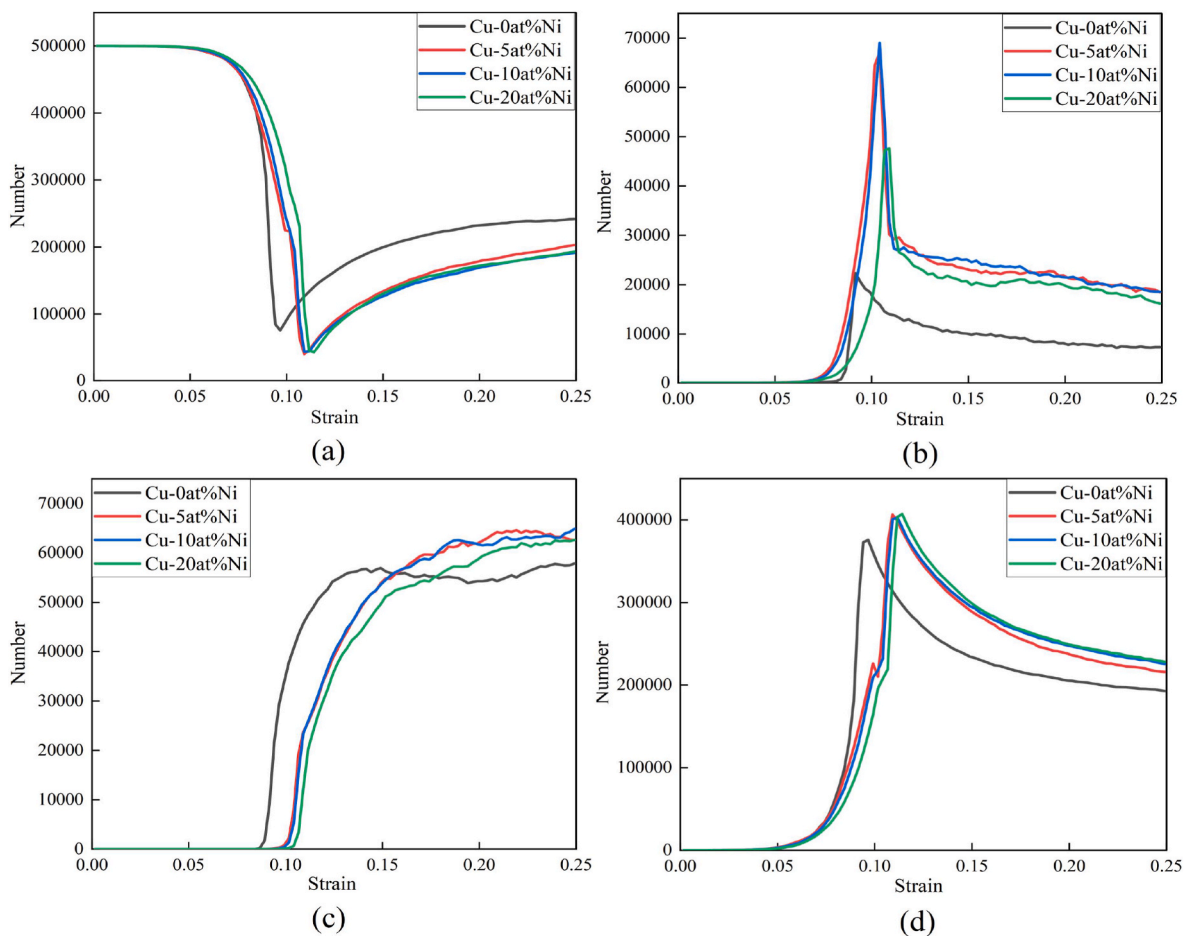


Fig. 17. Comparison of phase change structures of Cu and Cu–Ni alloys during the tensile process, where (a) FCC, (b) BCC, (c) HCP and (d) Other.

controlled. (2) Defects and impurities in real samples, such as dislocations and twins, stabilize the FCC structure and suppress phase transformation by pinning dislocations, reducing atomic rearrangements that could form BCC or HCP phases, whereas simulations neglect these imperfections, simplifying the material's response. (3) Macro-sized samples are influenced by environmental and size effects, which significantly affect material behavior, while simulations simplify these complexities. Larger samples experience strain gradients and boundary effects, absent in simulations. (4) MD simulations cover short time scales, usually nanoseconds to microseconds, while experiments involve long-term deformation, capturing processes like creep or recovery not fully addressed in MD simulations. In summary, these methodological differences—from idealized simulation conditions to the complexities of real-world experiments—explain the discrepancies in phase transformation and dislocation evolution in Cu–Ni alloys under strain.

3.3.2. Tensile properties of different PKA energies

To investigate the influence of irradiation defects on the mechanical properties of Cu–Ni alloys, we conducted a study focusing on the tensile properties of these alloys subjected to various PKA energies.

Fig. 20 presents the stress-strain curves for Cu–Ni alloys irradiated at different PKA energies along the [100] direction at a temperature of 300 K, accompanied by an enlarged view near the yield strength region. The results indicate that for Fig. 20(a) Cu system, the yield strength reaches its maximum at an irradiation energy of 0 keV, highlighting its superior mechanical properties. At 4 keV, the yield strength is the second highest, while at 10 keV, it is the lowest. For Fig. 20(b) Cu-5at%Ni system, the yield strength remains relatively stable at 2 keV when compared to the unirradiated state. However, a significant decrease in yield strength is

noted at 4 keV, with values at 6 keV falling between those observed at 2 keV and 4 keV. The yield strength shows minimal variation at 4 keV and 8 keV, but it drops markedly at 10 keV. In Fig. 20(c) Cu-10at%Ni system, the yield strength at 0 keV and 2 keV, as well as at 6 keV and 8 keV, are nearly equivalent, with the highest yield strengths recorded at 0 keV and 2 keV, and the lowest at 4 keV. For Fig. 20(d) Cu-20at%Ni system, the yield strength is highest at 0 keV, followed by 2 keV, with nearly identical yield strength observed at 4 keV, 8 keV, and 10 keV.

Table 2 summarizes the yield strength of Cu and Cu–Ni alloys after irradiation at 300 K with different PKA energies along the [100] direction. The data show that for the Cu system, the yield strength progressively decreases with increasing irradiation energy, peaking at 16.86 GPa at 0 keV and reaching a minimum of 15.82 GPa at 10 keV. The Cu-5at%Ni system exhibits a higher yield strength than the Cu system at the specified irradiation energies. Specifically, the yield strength decreases at 0 keV, 2 keV, and 4 keV, remains almost unchanged at 6 keV and 8 keV, and decreases again at 10 keV, with a minimum value of 17.05 GPa at 10 keV. In the Cu-10at%Ni system, the yield strength decreases at 4 keV, with the values at 4 keV and 6 keV being almost the same. The yield strength decreases again at 10 keV, reaching a minimum of 17.82 GPa at 10 keV. In the Cu-20at%Ni system, the yield strength decreases at 0 keV, 2 keV, 4 keV, and 6 keV, remains nearly unchanged at 8 keV, and decreases further at 10 keV. The minimum yield strength of 18.86 GPa occurs at 10 keV. These trends are illustrated in Fig. 22, where part (a) shows the relationship between yield strength and Ni content for Cu alloys under varying irradiation energies, while part (b) illustrates how yield strength changes with varying PKA energies for Cu–Ni alloys with different Ni contents (Cu, Cu-5at%Ni, Cu-10at%Ni, Cu-20at%Ni). Overall, the yield strength of Cu–Ni alloys increases with increasing Ni

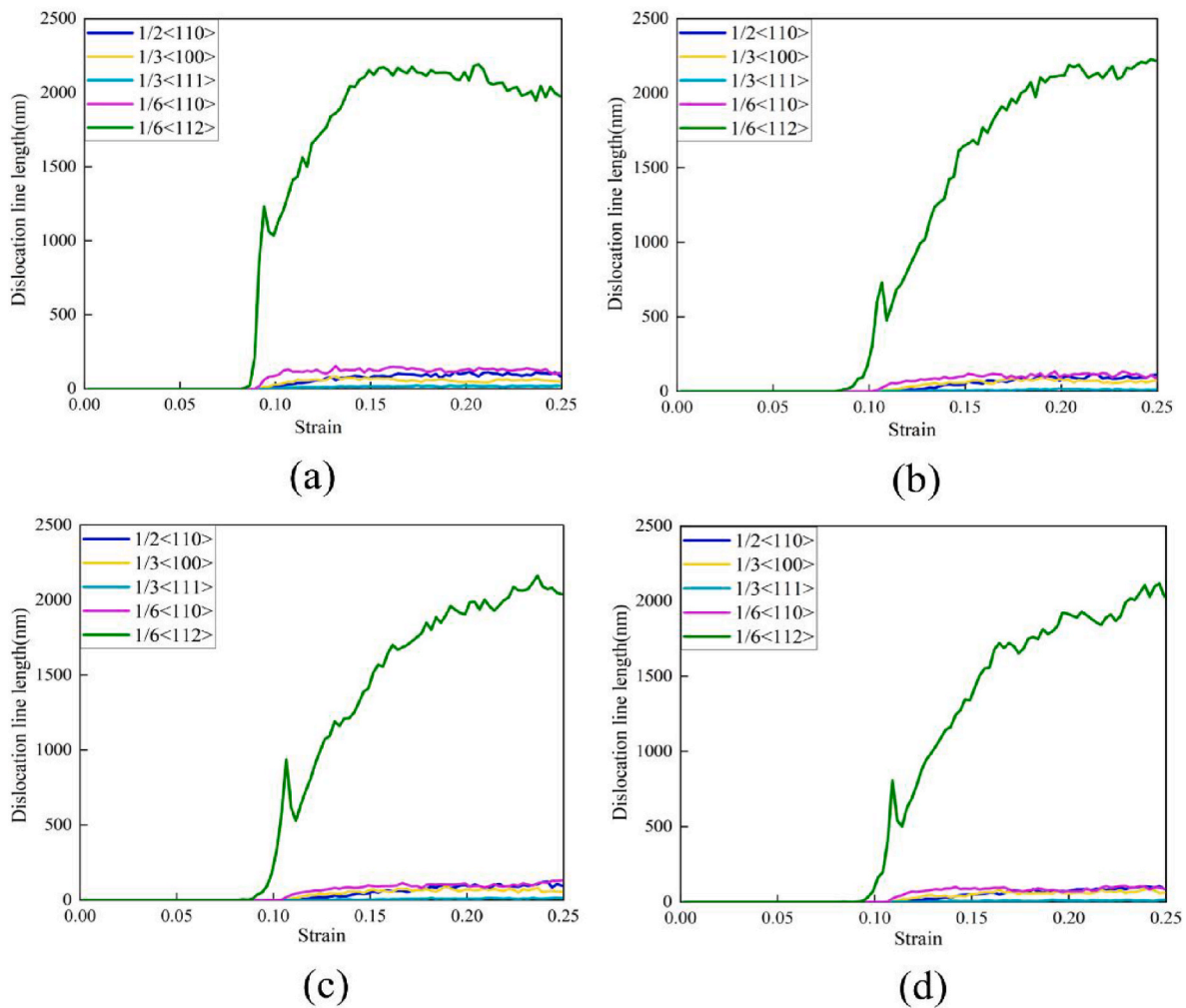


Fig. 18. The length of the dislocation line generated by (a) Cu-0at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni varies with strain during tensile process.

content and decreases with increasing irradiation energy.

In Fig. 21, we illustrate the temporal evolution of dislocation line lengths in Cu–Ni alloys subjected to irradiation at varying PKA energies. Our results demonstrate that dislocations are not detected within approximately 15 ps, which corresponds to the initial stretching phase of the crystal. In Fig. 21(a) Cu system, the dislocation line length is minimal at 50 ps when the PKA energy is 0 keV. Notably, the dislocation line lengths at 50 ps for PKA energies of 2 keV and 4 keV are nearly indistinguishable, as are those for 6 keV and 10 keV. Furthermore, the dislocation line lengths for PKA energies of 6 keV and 10 keV are significantly greater than those for 2 keV and 4 keV at the 50 ps mark, with the maximum length observed at 8 keV. In the case of Fig. 21(b) Cu-5at%Ni alloy, the dislocation line lengths remain relatively uniform at 50 ps across PKA energies of 0 keV and 2 keV, 4 keV and 6 keV, as well as 8 keV and 10 keV. For Fig. 21(c) Cu-10at%Ni alloy, a notable finding is that the dislocation line length peaks at a PKA energy of 4 keV. In Fig. 21(d) Cu-20at%Ni alloy, the dislocation line length reaches its maximum at a PKA energy of 10 keV. Additionally, a substantial increase in dislocation line length is observed with increasing strain.

4. Discussion

In the displacement cascade process, the process is divided into two distinct phases: the peak phase, which lasts until the thermal peak (characterized by rapid defect generation), and the stable phase following the thermal peak (during which defect recombination occurs). As the temperature increases, the number of Frenkel defect pairs in the

stabilization phase decreases, which is consistent with the enhanced atomic mobility at higher temperatures. The increase in temperature facilitates the recombination of interstitial atoms and vacancies, leading to a reduction in the number of defect pairs during the stabilization phase. Another possible explanation for this phenomenon is that higher temperatures shorten the displacement collision sequence in the cascade, leading to a more compact defect configuration. This compact configuration likely reduces the distance between vacancies and interstitial atoms, promoting their recombination, a finding consistent with previous studies [52,65,66]. Unlike previous studies, which typically focus on the effects of temperature changes within a narrow range, this study further expands the understanding of temperature's influence on defect evolution by considering different Ni contents and a broader temperature range. Our findings show that as PKA energy increases, the energy released in collisions between PKA atoms and lattice atoms also increases. This increase in energy intensifies the cascade collisions, causing more lattice atoms to be excited and displaced, resulting in more interstitial atoms and vacancies. As the number of collisions increases, more Frenkel defect pairs are generated. Due to high-energy collisions promoting the migration of more defect atoms and the formation of new defects, a large number of Frenkel defect pairs are produced at the peak stage, ultimately increasing the number of defect pairs in the steady state. This observation is consistent with previous reports [66,67]. Additionally, displacement cascades in Cu–Ni alloys exhibit anisotropic irradiation characteristics, primarily due to variations in atomic periodicity and spacing within the crystal structure, especially across different crystal orientations. Similar studies have been conducted on

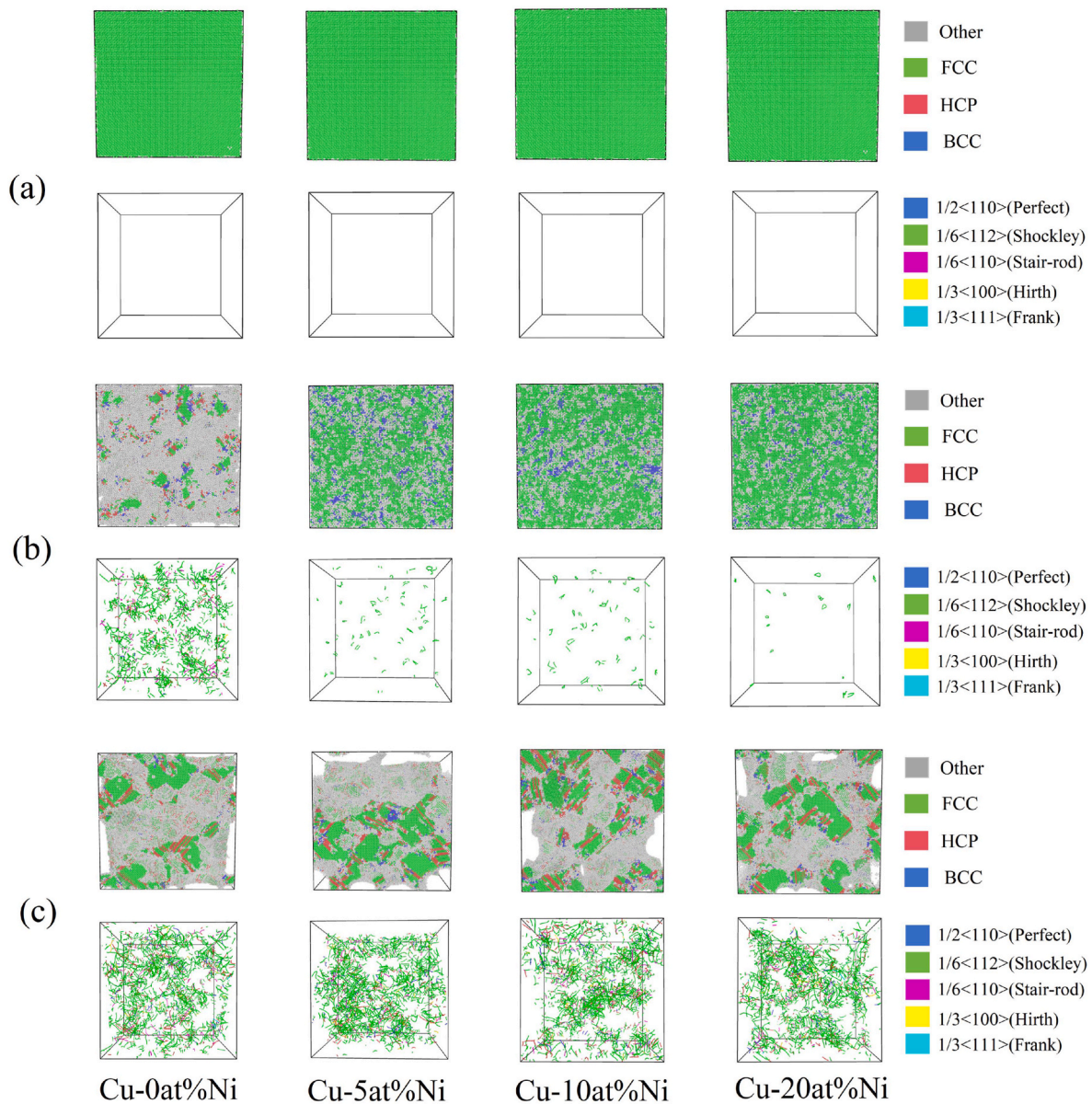


Fig. 19. The Microstructural diagrams of phase structure and dislocations produced in Cu and Cu–Ni alloys during tensile stretching, where (a) strain = 0.02 (b) strain = 0.09 (c) strain = 0.25.

HCP-Ti in the PKA energy range of 0.3–5.0 keV [68]. However, these studies only investigated the effects of crystal orientation in HCP-structured titanium (PKA energy range of 0.3–5.0 keV). In contrast, our study focuses on Cu–Ni alloys with an FCC structure and selects a higher PKA energy range to investigate the effect of crystal orientation. The results show that this phenomenon becomes more pronounced at higher PKA energies. This is because collisions between high-energy particles and material atoms disrupt the original lattice structure and generate more defects, intensifying atomic arrangement differences in different directions.

Irradiation can significantly alter the microstructure of materials, thereby affecting their mechanical properties. This study investigates the mechanical behavior of Cu–Ni alloys before and after irradiation. The results show that all alloys undergo three stages: elastic deformation, yielding, and localized deformation, as documented in Refs. [69, 70]. Among these Cu–Ni alloys, the yield strength increases with the nickel content. This enhancement is attributed to the role of nickel in promoting the formation of BCC, HCP, and other phase structures during tensile testing, while the dislocation line length decreases with

increasing nickel content. Similar phenomena have been observed in molecular dynamics tensile simulations of other materials, as reported in Refs. [69–71]. Notably, this study innovatively observes the changes in the mechanical properties of Cu–Ni alloys before and after irradiation through tensile simulations. Comparative results show that the yield strength of the irradiated alloys significantly decreases compared to their unirradiated counterparts, with a gradual decline in yield strength as the PKA energy increases. This trend can be explained by the higher PKA energy during irradiation, which promotes the formation of Frenkel defect pairs. These defects accumulate and evolve under tensile stress, leading to the aggregation of dislocations and other defects. This process induces crystal slip or twinning, resulting in a significant increase in dislocations and premature yielding of the material. Therefore, the decrease in tensile strength is closely related to the sharp drop in stress during the strain process. From the comparison in Tables 2 and it is evident that, under 10 keV irradiation, the yield strength of irradiated materials is generally lower than that of unirradiated materials. Specifically, the yield strength of Cu-5at%Ni alloy decreased by 1.86 GPa, the yield strength of Cu-10at%Ni alloy decreased by 1.45 GPa, and the

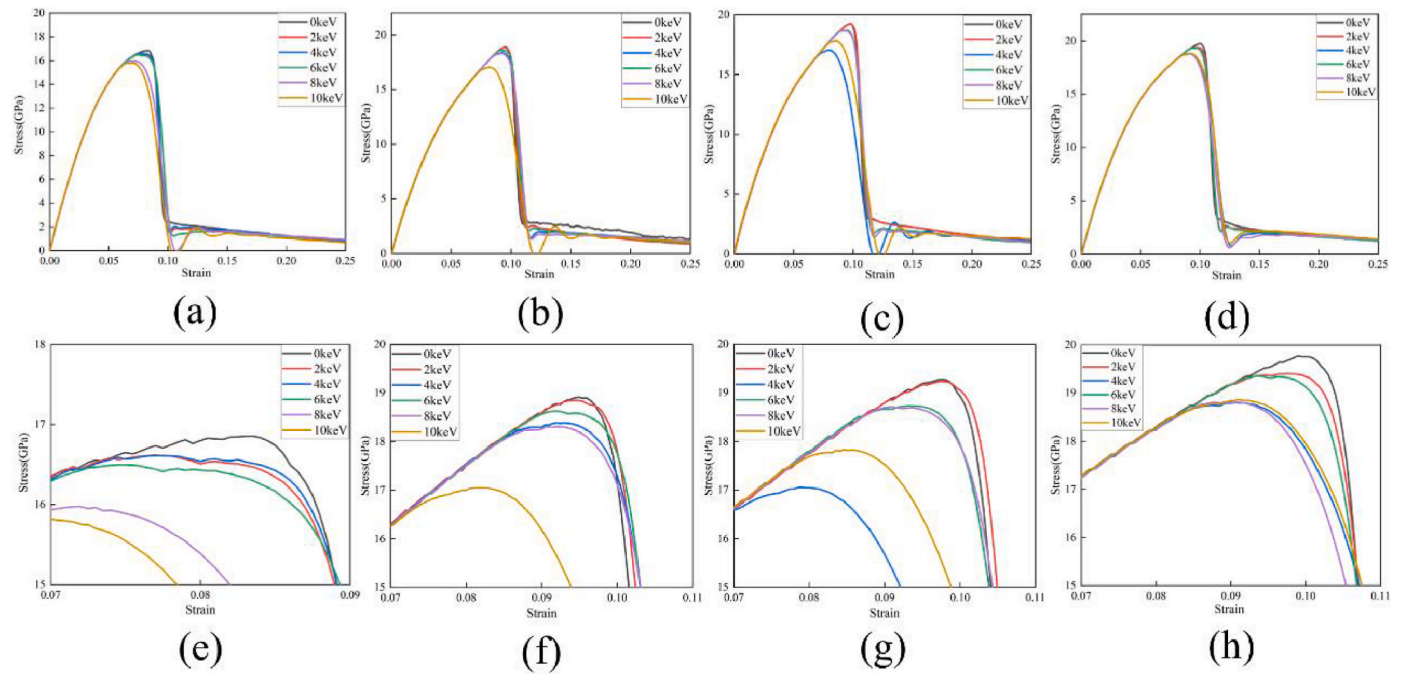


Fig. 20. Stress-strain curves of (a) Cu-0at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni after irradiation with different PKA energies in the direction [100] at 300 K, (e),(f),(g) and (h) are amplified corresponding graphs near the yield strength in (a), (b), (c) and (d), respectively.

Table 2

Yield strength (GPa) of Cu and Cu–Ni alloys after irradiation at 300 K with different PKA energies along the [100] direction.

Alloys	0 keV	2 keV	4 keV	6 keV	8 keV	10 keV
Cu-0at%Ni	16.86	16.62	16.61	16.49	15.97	15.82
Cu-5at%Ni	18.91	18.85	18.38	18.32	18.31	17.05
Cu-10at%Ni	19.27	19.23	18.66	18.46	18.41	17.82
Cu-20at%Ni	19.77	19.41	19.10	18.98	18.91	18.86

yield strength of Cu-20at%Ni alloy decreased by only 0.91 GPa. Moreover, the yield strength of Cu-5at%Ni alloy before irradiation was 18.91 GPa, while the yield strength of Cu-20at%Ni alloy was 19.77 GPa, which is significantly higher than that of the alloys with lower nickel content. These results clearly indicate that increasing nickel content indeed enhances the irradiation resistance of the material.

5. Conclusion

This study systematically analyzes the displacement cascade process and mechanical properties of Cu–Ni alloys under different temperatures, PKA energies, and crystal orientations using molecular dynamics simulations. The results show that, with PKA energy and crystal orientation held constant, temperature significantly influences defect formation. Specifically, as the temperature increases, the number of Frenkel defect pairs in Cu–Ni alloys decreases, mainly due to high temperature promoting the recombination of interstitial atoms and vacancies, thereby reducing the number of defect pairs in the stable phase. Meanwhile, with increasing PKA energy, the number of defects increases significantly, which can be attributed to higher collision energy intensifying cascade collisions, further promoting defect formation. Additionally, when temperature and PKA energy are held constant, the PKA orientation also significantly affects the spatial distribution of defects. By analyzing the combined effects of temperature, PKA energy, and crystal orientation, this study reveals how these factors collaboratively impact the microstructure and macroscopic mechanical properties of Cu–Ni alloys.

Regarding the effect of Ni content, the study shows that as Ni content increases, the number of Frenkel defect pairs in Cu–Ni alloys increases

until Ni concentration reaches 5 at%. When the Ni concentration exceeds this threshold, the number of defects decreases significantly, especially when Ni content is higher (e.g., 20 at%). This suggests that the addition of Ni enhances the formation and recombination efficiency of defects in the Cu alloy, effectively reducing the number of defects. However, under 300 K irradiation conditions, when Cu–Ni alloys with different Ni contents are irradiated with varying PKA energies, the alloys exhibit brittle behavior, which is closely associated with higher PKA energies. Increased PKA energy promotes the formation of more Frenkel defect pairs, further increasing dislocation generation, thereby exacerbating the material's brittleness. Further tensile simulation tests show that, as Ni content increases, the yield strength of Cu–Ni alloys increases significantly, particularly when Ni content reaches 20 at%, where the strength enhancement is particularly notable. After irradiation, the incorporation of Ni plays a crucial role in maintaining the alloy's strength. Cu–Ni alloys with higher Ni content (e.g., 20 at%) exhibit significantly better mechanical properties before and after irradiation compared to alloys with lower Ni content (e.g., 5 at%), and in terms of irradiation resistance, alloys with higher Ni content show less performance degradation. This indicates that Ni incorporation not only enhances the strength of the alloy but also improves its stability under irradiation conditions.

In conclusion, the number of Frenkel defect pairs has a significant impact on the mechanical properties of Cu–Ni alloys. Furthermore, the study finds that Cu–Ni alloys with higher Ni content (20 at%) exhibit superior performance in both mechanical and irradiation resistance properties. Further research will explore whether increasing Ni content can further enhance irradiation performance, evaluating its potential to improve both mechanical and irradiation resistance properties, and further validate its potential for performance enhancement. Overall, this study provides a promising strategy for designing materials with improved mechanical properties and irradiation resistance, particularly in selecting the optimal Ni content for Cu–Ni alloys. This work offers valuable insights for developing materials that can withstand extreme environmental conditions and contributes to the advancement of radiation-resistant technologies.

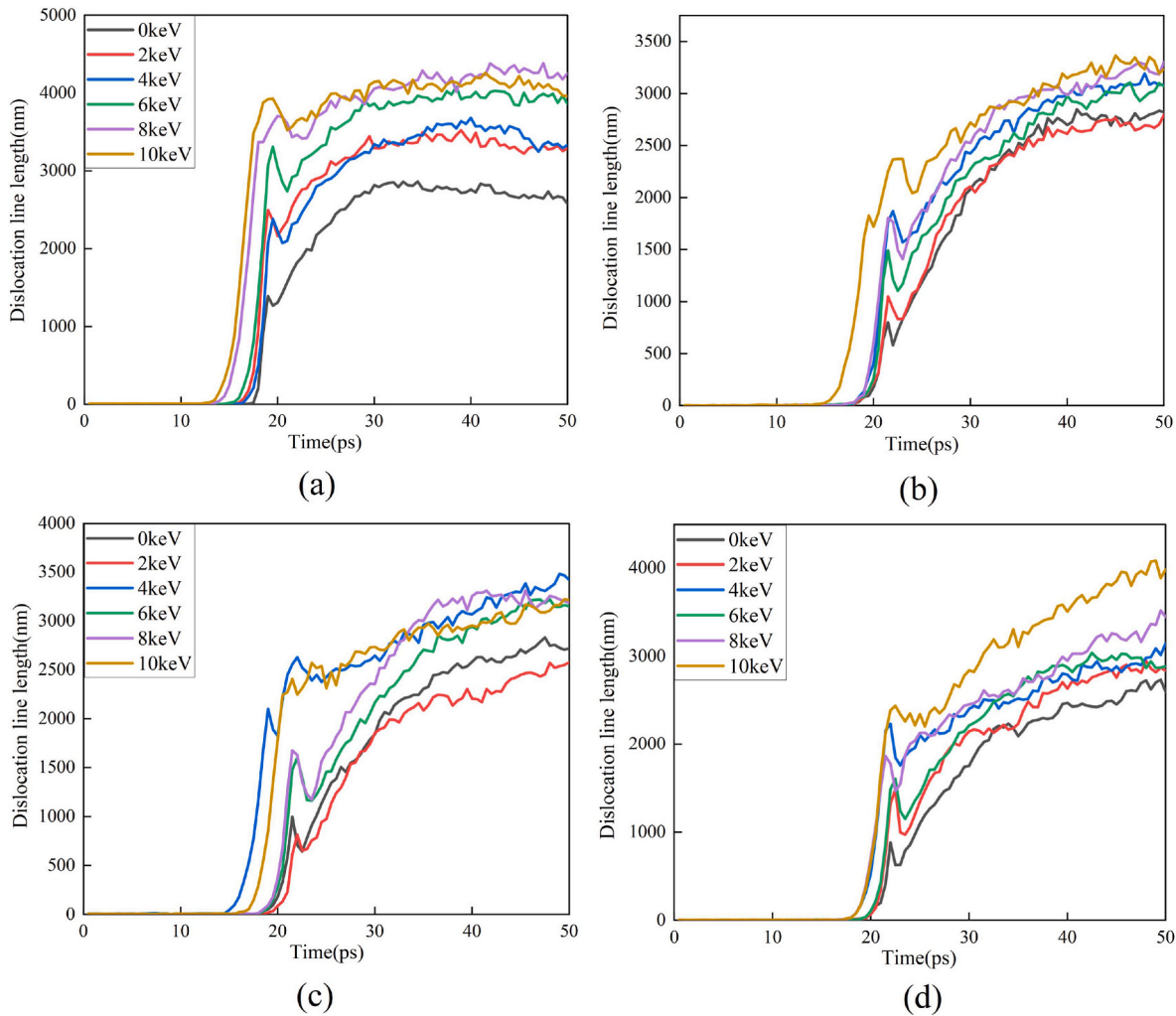


Fig. 21. The lengths of dislocation lines during the stretching of (a) Cu-0at%Ni, (b) Cu-5at%Ni, (c) Cu-10at%Ni and (d) Cu-20at%Ni under different PKA energy irradiations as a function of time.

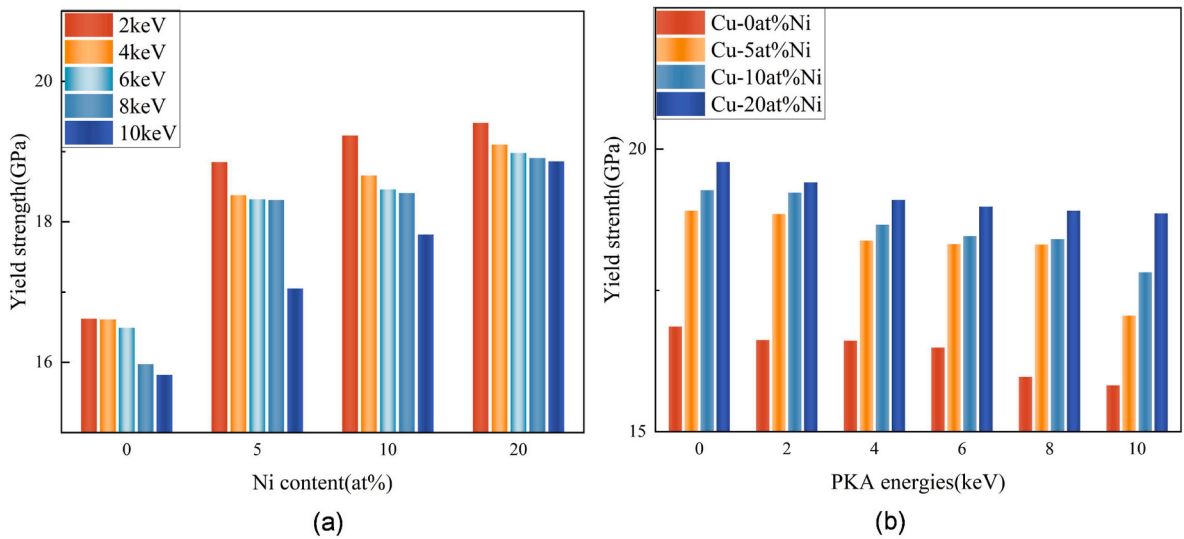


Fig. 22. Yield strength (GPa) of Cu and Cu-Ni alloys after 300 K irradiation with different PKA energies along the [100] direction. (a) Stress vs. Ni content for varying irradiation energies. (b) Stress vs. PKA energy for Cu alloys with different Ni contents.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A: List of Abbreviations

Abbreviation	Meaning	Explanation
PKA	Primary Knock-on Atom	In irradiation, it refers to the first atom displaced from its position in a material when struck by a high-energy particle. It initiates subsequent atomic displacements, leading to damage in the material.
MD	Molecular Dynamics	A simulation technique used to model atomic and molecular movements in materials.
DFT	Density Functional Theory	A computational method used to investigate the electronic structure of atoms, molecules, and solids.
VASP	Vienna Ab-initio Simulation Package	A software package based on density functional theory (DFT) for performing first-principles calculations. https://www.vasp.at
LAMMPS	Large-scale Atomic/Molecular Massively Parallel Simulator	A powerful parallel simulation package used for atomic-level molecular dynamics simulations. https://www.lammps.org
CG	Conjugate Gradient	A numerical optimization method used for minimizing energy in simulations.
FCC	Face-Centered Cubic	A crystal structure where atoms are located at the corners and center of each face of a cube.
BCC	Body-Centered Cubic	A crystal structure with atoms at the center and corners of the cube.
HCP	Hexagonal Close-Packed	A crystal structure where atoms are arranged in a hexagonal pattern.
ZBL	Ziegler-Biersack-Littmark Potential	A potential model used to describe interactions between atoms during radiation damage.
NRT	Number of Particles, Pressure, and Temperature Ensemble	In molecular dynamics simulations, it is used to model systems under isothermal and isobaric conditions.
NVE	Constant Number of Particles, Volume, and Energy	A simulation ensemble in which the number of particles, volume, and energy are kept constant.
W-S	Wigner-Seitz	A method for identifying atomic defects in materials based on the position of atoms within a unit cell.
OVITO	Open Visualization Tool	Software used for the visualization and analysis of atomistic simulation data https://www.ovito.org
DXA	Dislocation Analysis	A method for identifying and analyzing dislocations within a material.
CNA	Common Neighbor Analysis	A method for identifying atomic configurations to classify defects in crystalline materials.
EAM	Embedded Atom Model	A model used to simulate interactions between atoms, commonly used for metals.

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